

ENT BY DR SHUBHAM MITTAL

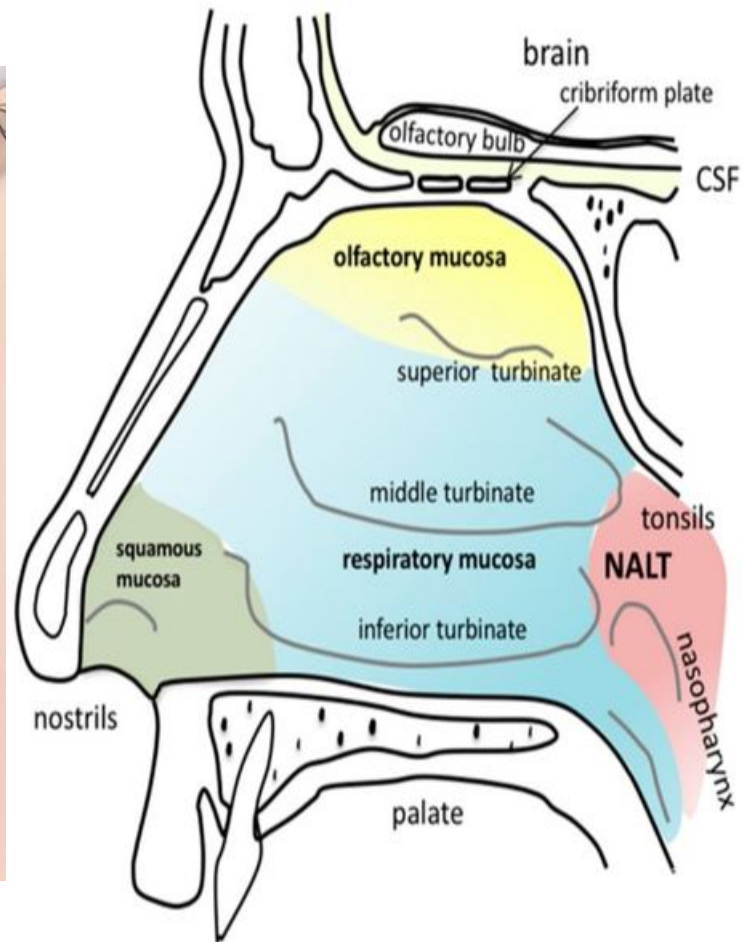
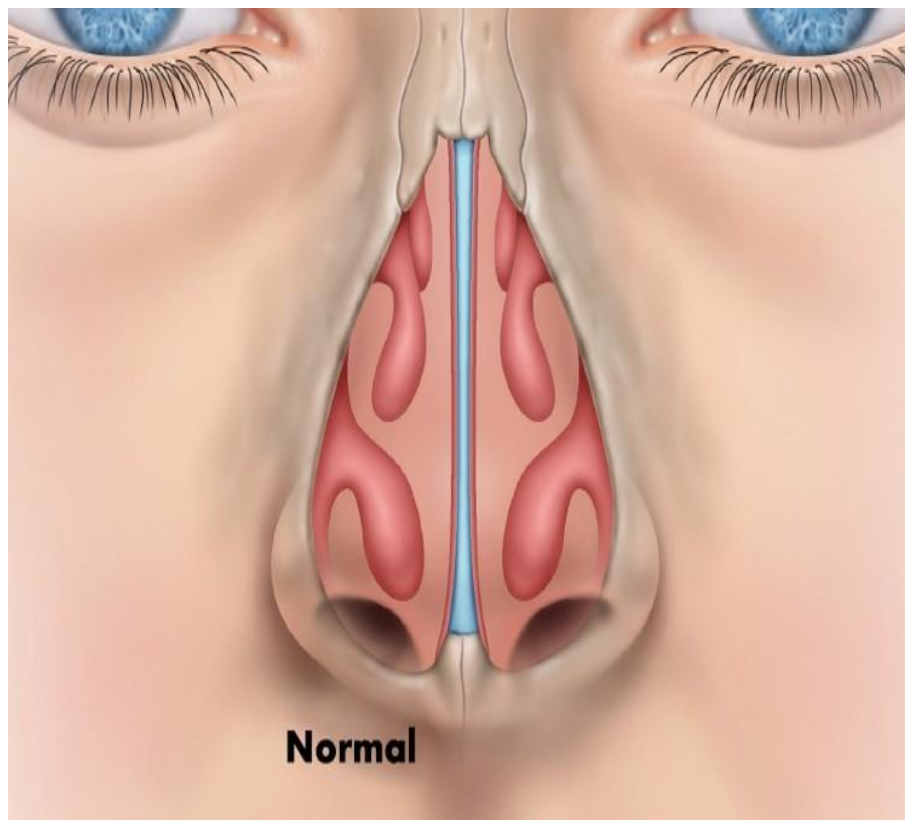


ANATOMY OF NOSE



1. UPPER 1/3RD BONY, LOWER $\frac{2}{3}$ CARTILAG
2. ALL TURBINATES ARE PART OF ETHMOID EXCEPT IT
3. LARGEST -IT
4. PH OF NOSE 7
5. CHOANA IS A JUNCTION NOSE AND NASOPHARYNX
6. AP LENGTH, NASAL SEPTUM -8CM

1. UPPER 1/3RD BONY, LOWER $\frac{2}{3}$ CARTILAGENOUS
2. ALL TURBINATES ARE PART OF ETHMOID BONE EXCEPT IT
3. LARGEST -IT
4. PH OF NOSE 7
5. CHOANA IS A JUNCTION NOSE AND NASOPHARYNX
6. AP LENGTH, NASAL SEPTUM -8CM



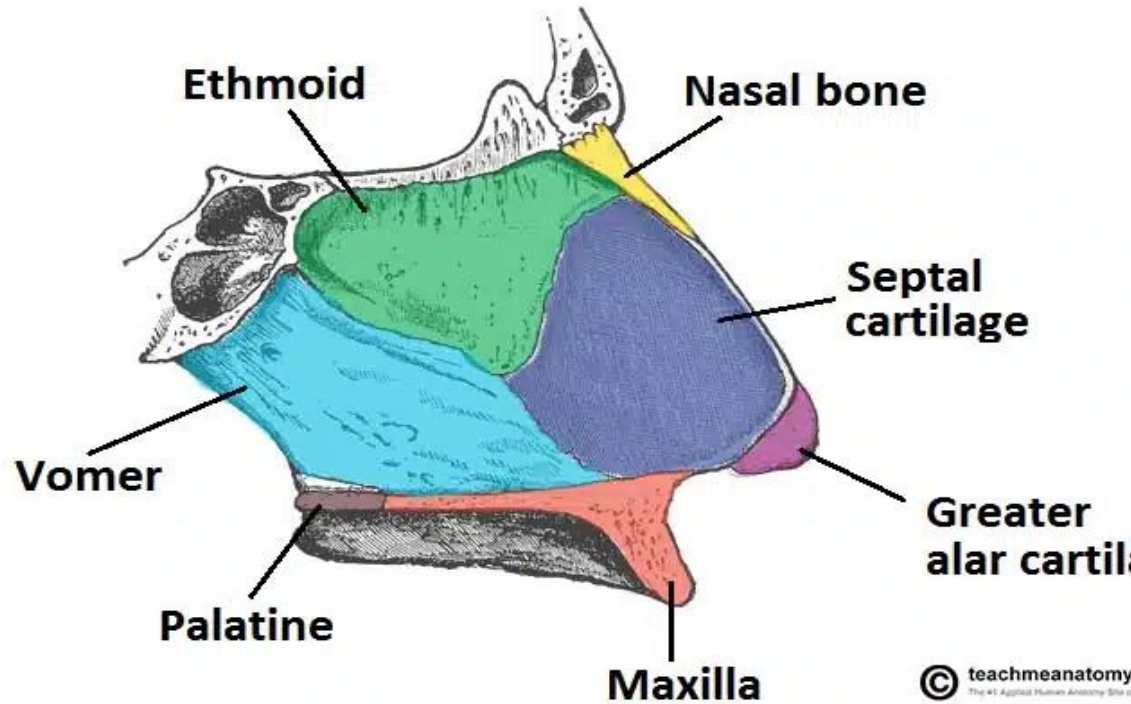
MINOR PARTS OF SEPTUM

SPINE OF FRONTAL BONE

SPINE OF MAXILLA

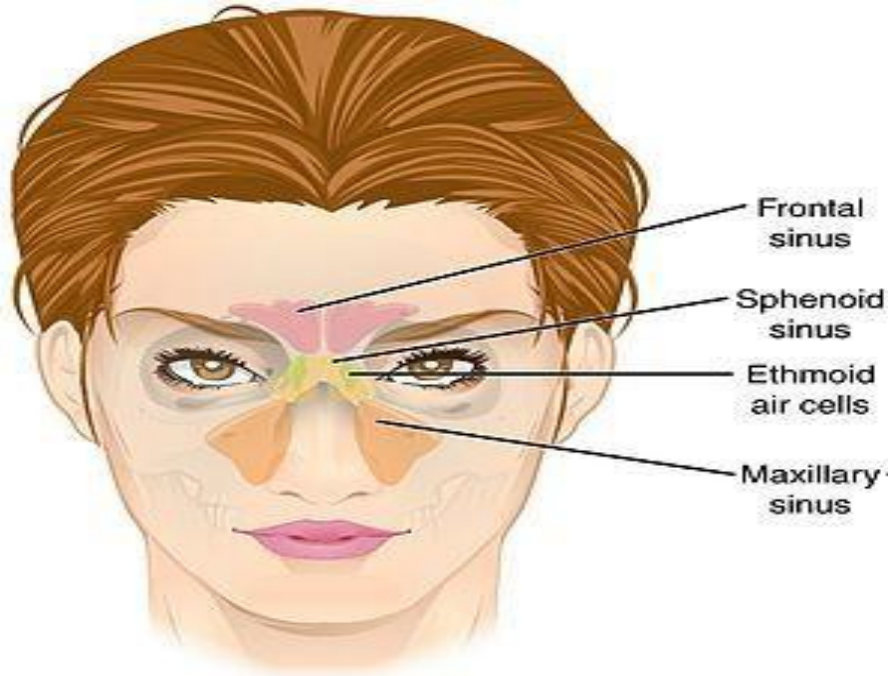
PALATINE BONE

ROSTRUM OF SPHENOID



MAJOR PARTS OF NASAL SEPTUM

PNS



1. Maxillary sinus (antrum of highmore)
 - Largest sinus
 - Below the orbit
 - Volume: 15ml
2. Frontal sinus
 - Just above eyebrows.
 - Above the orbit
3. Sphenoid sinus
4. Ethmoid sinus (ethmoid air cells)
 - Between 2 orbit

LARGEST SINUS-15 ML

MOST FUNCTIONAL PNS-15 AIR CELL

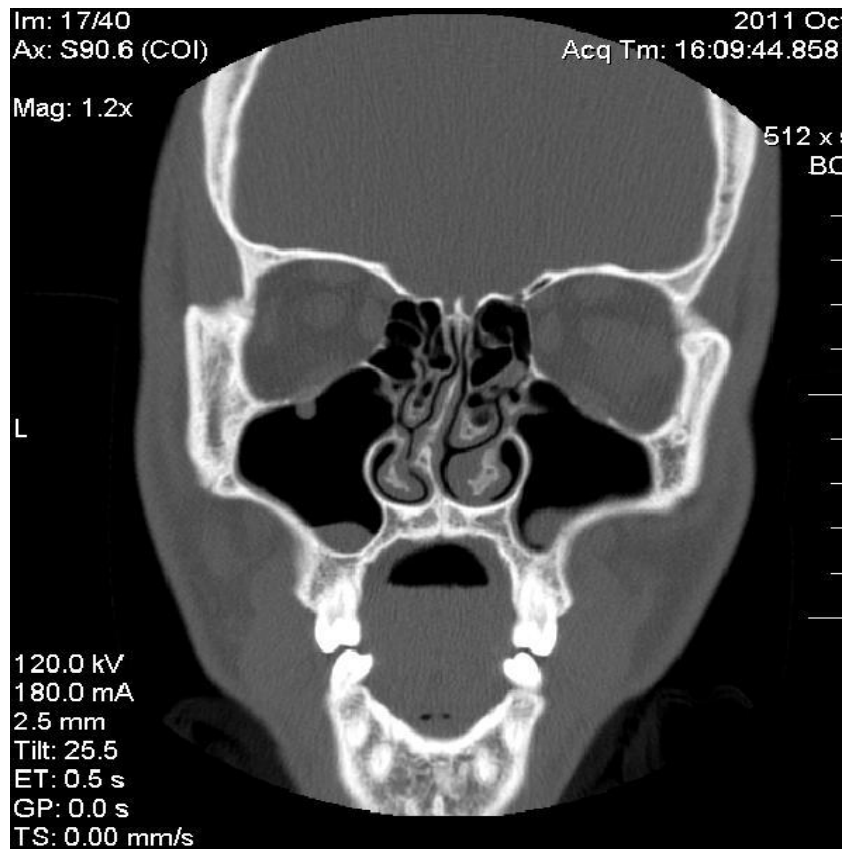
AGGER NASI

BULLA ETHMOIDALIS

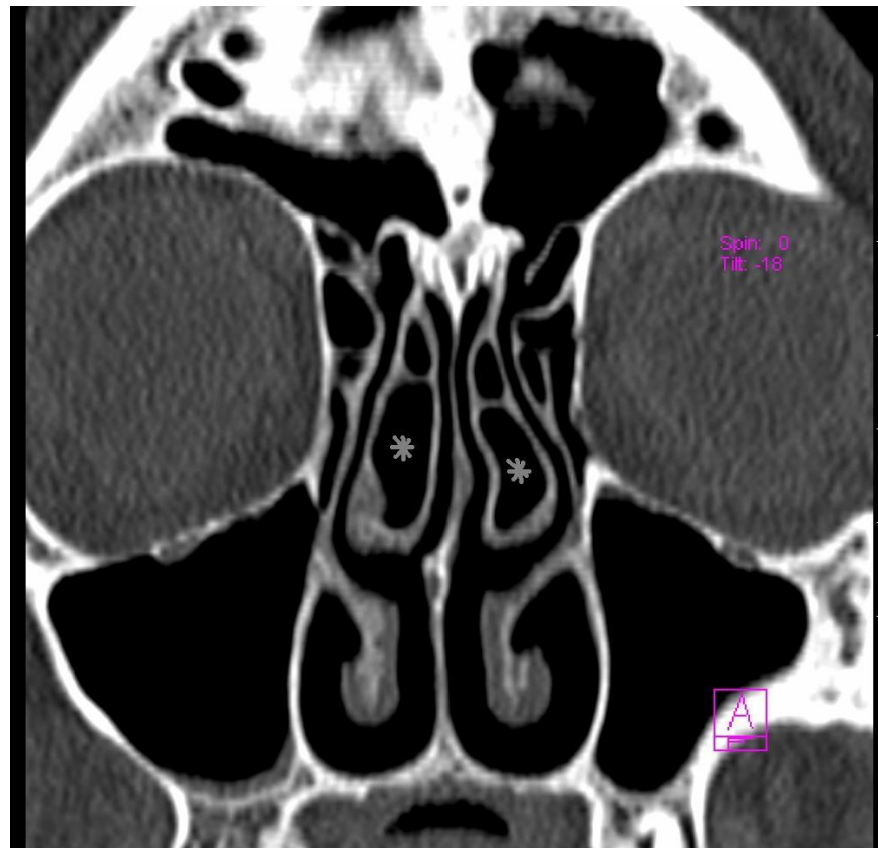
ROOF OF MAXILLARY SINUS-FLOOR OF ANT. ETHMOID+ORBITAL FLOOR
:HALLER CELL

MOST POSTERIOR-ONODI-ON

CT SCAN NORMAL



CHONCHA BULLOSA



GLABELLA- MOST PROMINENT PART FRONTAL BONE

RADIX-UPPER EDGE NOSE

NASION- NASO FRONTAL ANGLE-134 DEGREE

STRUCTURES IN MEATUS

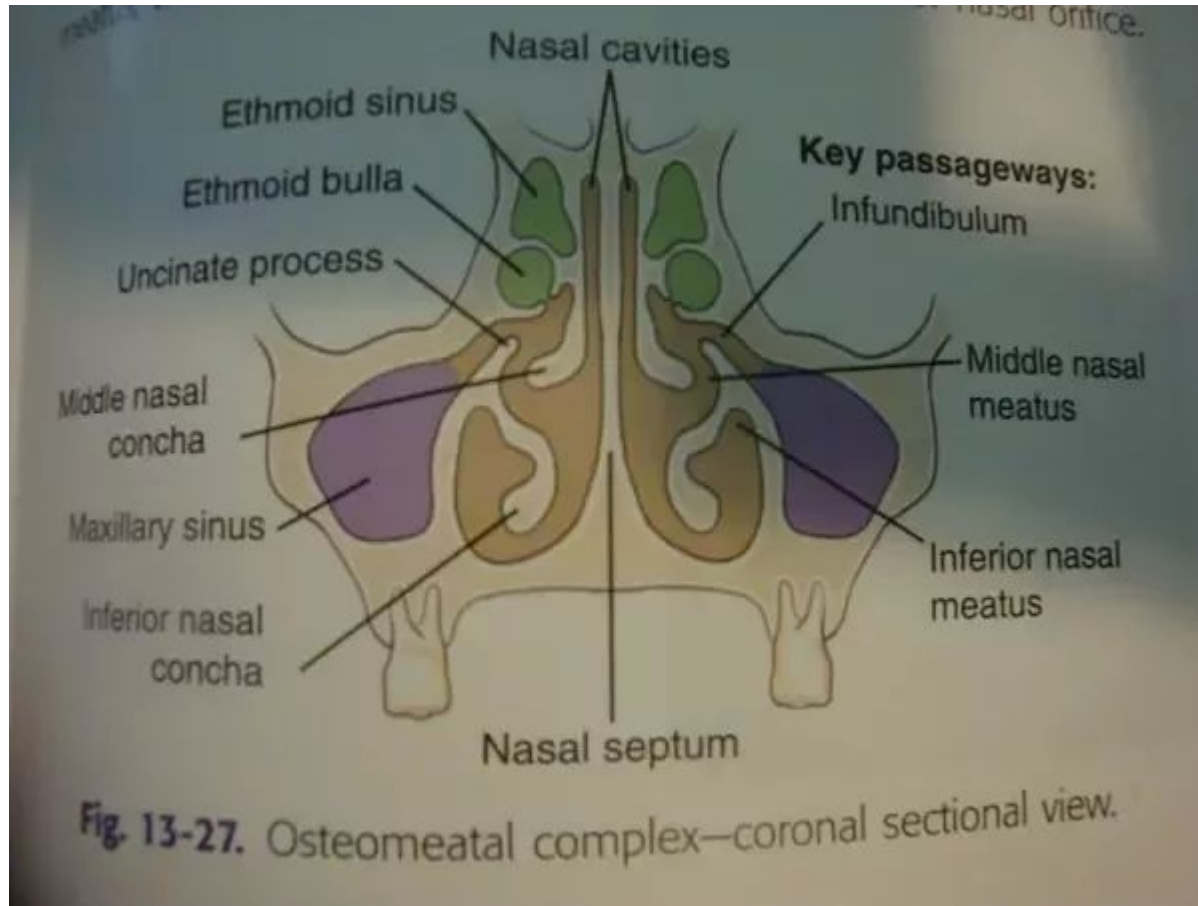
INF MEATUS

MIDDLE MEATUS- Maxillary, Frontal, Ant Ethmoid

SUPERIOR MEATUS- post ethmoid opens here

SPHENOETHMOIDAL RECESS- sphenoid sinus opens here

OSTEOMEATAL COMPLEX



-APPEARANCE OF SINUSES- MESF

MAXILLARY

FRONTAL- DOES NOT DEVELOP UNTILL 4 YR OLD

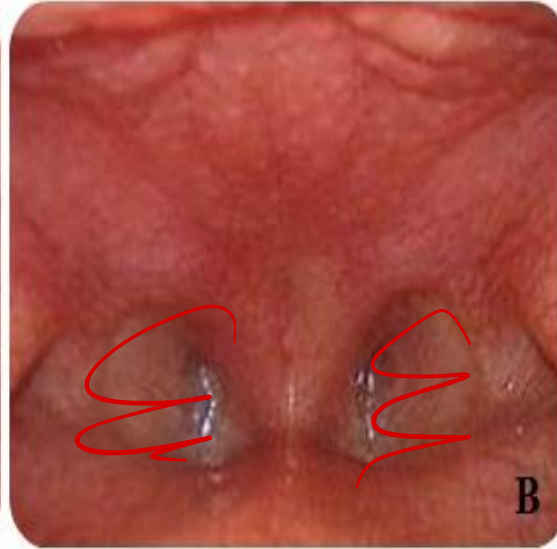
SPHENOID

ETHMOID - 1YEAR

CHOANA

CHOANAL ATRESIA

- CHOANAL ATRESIA
- B/L CHOANAL ATRESIA- CYANOSED BLUE BABY TURNS PINK ON CRY.



1. Choanal + $\Rightarrow$ Persistence
of
Bucco nasal
membrane

2. C/A $\rightarrow$ Difficulty in Breathing

Sign $\rightarrow$ Cyanosis

Dx $\rightarrow$ N.H tube won't pass

$T/t \Rightarrow$ 1. Cyanosis disappears
as soon as the child
cries.

2. Mc Govern's Nipple.

3. Endoscope - later.

✓ C/C $\Rightarrow$ difficulty in breathing

Sign $\Rightarrow$ Cyanosis

Dx $\Rightarrow$ Ryles tube won't pass.

T/t $\Rightarrow$ 1. Cyanosis disappears
as soon as child chills.

2. ~~Mc~~ Govern's
Nipple.

3. Laser & Endoscopic
Sx.

MUCORMYCOSIS



- it is fungal infection of nose by mucor group of fungi.
- It is seen in young, diabetic, HIV positive patient. (immuno compromise pts)

Clinical feature:

Mucor is angioinvasive fungus – it enter to orbit and brain –

1. Blackish mass in nose

blackish discoloration around eye Rx: DOC:
amphotericin-B

[Liposomal]

SINUSITIS

VIRAL -RHINO

BACTERIAL-STREPTO

FUNGUS-ASPERGILLUS FUMIGATUS

MORE IN NOSE

NIGER MORE IN EAR

SINUS INVOLVEMENT- M > F > E > S

PAEDS- E > M > F > S

Nephrotoxic [↓ K⁺]
Hypokatemia

c/f .- ANOSMIA

BLOCKAGE

CONGESTION

DISCHARGE

FACIAL PAIN ,FEVER AND HALITOSIS

IOC- DIAGNOSTIC ENDOSCOPY

RADIOLOGICAL IOC-CT

TOC-FESS

SWELLINGS

CHEECK/ DENTAL/LOWER EYELID

ROOT/RADIX/DORSUM/UPPER AND LOWER EYELID

OFFICE HEADACHE/FRONTAL HEADACHE/MORE IN MORNING LESS IN
EVENING

RETRO-ORBITAL/OCCIPITAL/VISION

COMPLICATIONS

FRONTAL SINUS: MUCUS-----> PUS----->OSTEOMYELITIS

A/K/A POTTS PUFFY TUMOUR

ORBITAL ABSCESS- ETHMOID SINUS



ALLERGIC FUNGAL RHINOSINUSITIS

CHARACTERISTIC QUESTION ASKED

1. PEANUT BUTTER APPEARANCE
2. X-RAY/CT DOUBLE DENSITY



BENT KUHN CRITERIA

MAJOR

1. ALLERGIC TYPE 1 HYPERSENSITIVITY REACTION EVIDENCE
2. NASAL POLYPS
3. CT FINDING
4. EOSINOPHILIC MUCUS
5. POSITIVE FUNGAL SMEAR

MINOR

1. ASTHMA
2. UNILATERAL PREDOMINANCE(DISEASE)
3. BONE EROSION IN RADIOGRAPHS
4. POSITIVE FUNGAL CULTURE
5. CHARCOT LEYDEN CRYSTALS
6. EOSINOPHILIA IN SERUM

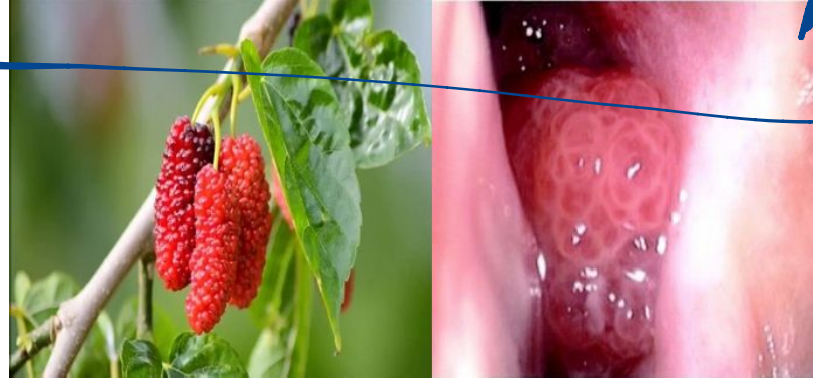
RHINITIS

ALLERGIC RHINITIS

HYPERTROPHIC RHINITIS

- MULBERRY LIKE NASAL MUCOSA (HALLMARK)

T/T - KTP LASER



Rhinosporidiosis >> Hyp. Rhinitis

Rhinosporidiosis
H.R

- DRUG INDUCED RHINITIS

VASODILATORS LIKE

1. ANTIHYPERTENSIVES
2. OCPS

CAUSE IT

T/T - VASOCONSTRICTORS- OXYMETAZOLINE

- RHINITIS MEDICAMENTOSA
- VASOMOTOR RHINITIS - Vidian Neurectomy done to treat it



ATROPHIC RHINITIS

HERNIA

MERCIFUL ANOSMIA

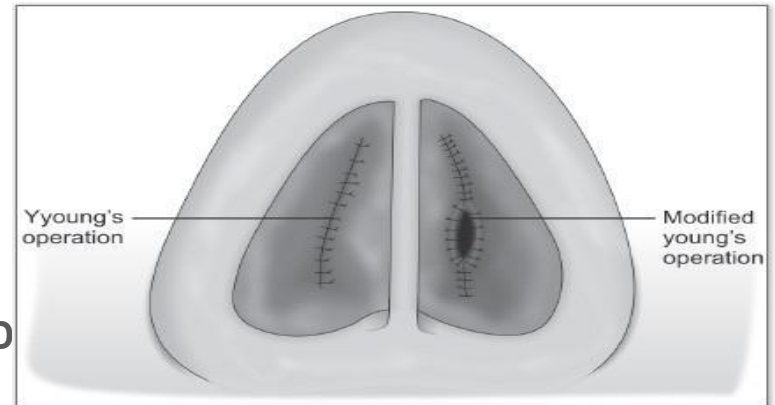
ALKALINE NASAL DOUCHE

SURGERY- YOUNG OPERATION: BLOCK NOSE
FOR 6 MONTHS

MODIFIED YOUNG: PARTIAL CLOSURE

INJ. TEFLON TO LATERAL WALL

LAUTENS LASER SURGERY -LATEST NEW METHOD



NASAL BONE FRACTURE

REDUCE FRACTURE IMMEDIATELY(CLOSED REDUCTION)- IF NO EDEMA

IF OEDEMA PRESENT—>ANTIINFLAMMATORY FOR 5 DAYS—>REDUCTION

- REDUCTION IF DONE IN LESS THAN 3 WEEKS - DO CLOSED REDUCTION
- MORE THAN 3 WEEKS - DO OPEN REDUCTION

Asch septal forceps



Fig. : Asch septal forceps

Walsham forceps



Fig. : Walsham forceps

Asch forceps

Walsham forceps

ZYGOMATIC FRACTURE- also called as tripod fracture

LE FORTE FRACTURE

1ST

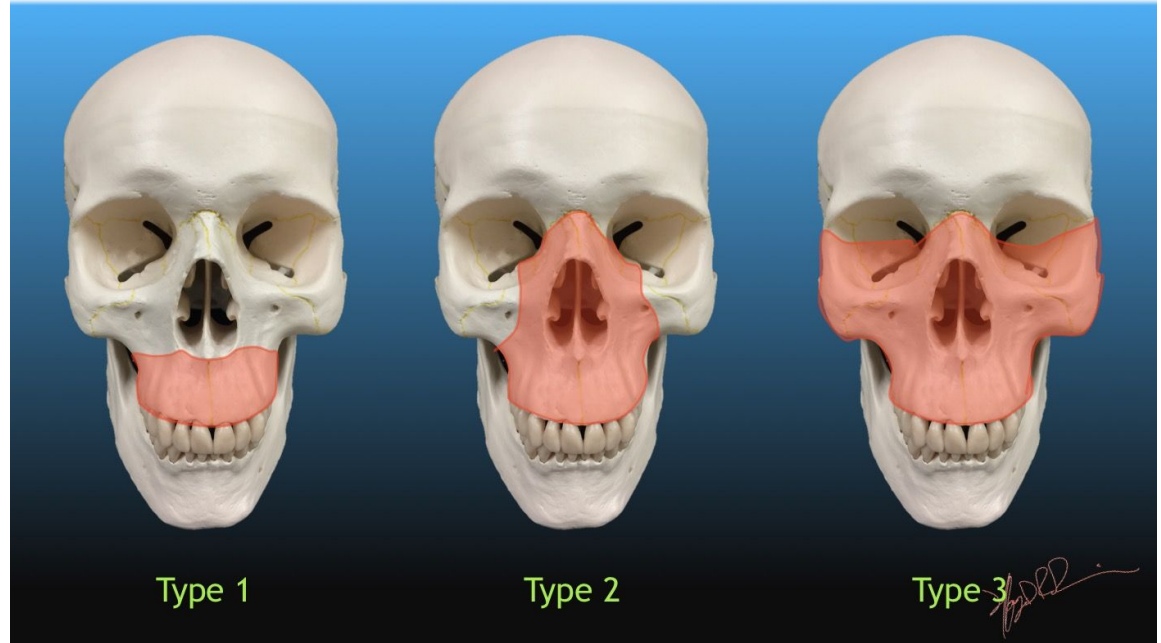
- LOW MAXILLARY FRACTURE
- FLOATING PALATE
- GUIREN SIGN

2ND

PYRAMIDAL FRACTURE

3RD

CRANIOFACIAL FRACTURE



- LE FORTE 2 AND LE FORTE 3 ARE ASSOCIATED WITH CSF RHINORRHEA

DIRECTION OF TRAUMA

CLASS

- 1 CHEVALLET - TRAUMA FOR BELOW- VERTICAL #
- 2 JARJAVAY- TRAUMA FRONT/FRONTOLATERAL-HORIZONTAL #
- 3 MULTIDIRECTIONAL

CSF RHINORRHEA

TRAUMA TO CRIBRIFORM PLATE (MORE PRECISELY FOVEA ETHMOIDALIS)

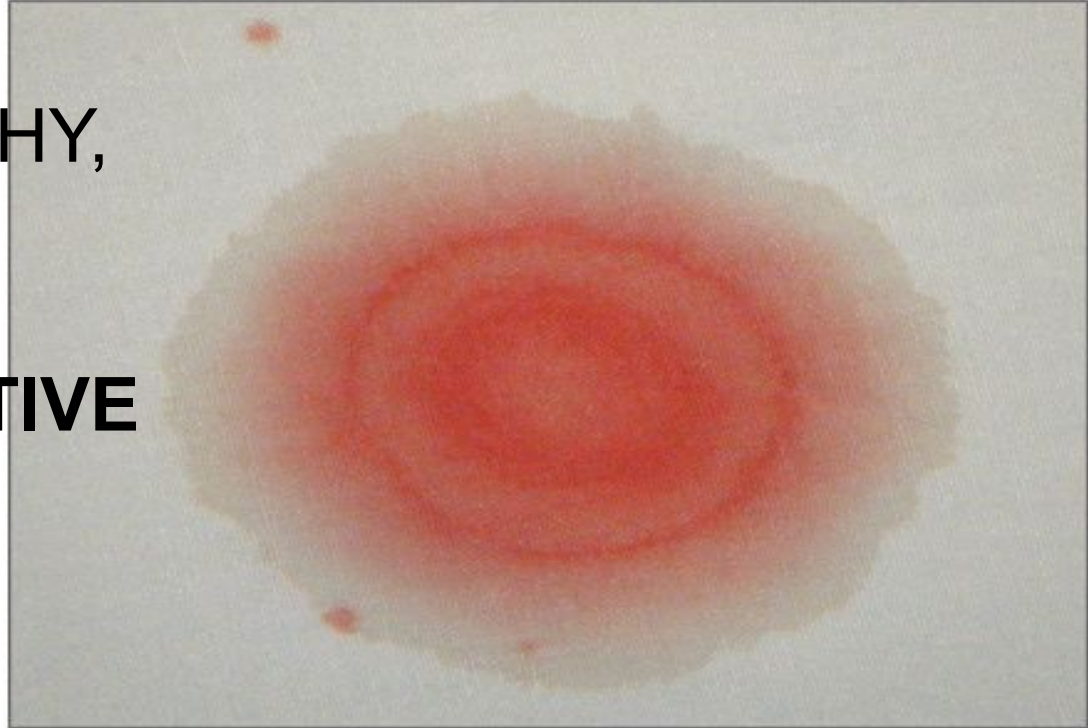
90% TRAUMATIC- M/C SURGICAL TRAUMA (IATROGENIC LIKE FESS SURGERY)

DOUBLE RING/TARGET/HALO SIGN- TRAUMATIC



BETA 2 TRANSFERRIN TEST

WE CAN DO MRI ,
MR CISTERNOGRAPHY,
HRCT ALSO
MANAGEMENT IS
MOSTLY CONSERVATIVE



BETA 2 TRANSFERRIN TEST

WE CAN DO MRI , MR CISTERNOGRAPHY, HRCT ALSO

MANAGEMENT IS MOSTLY CONSERVATIVE

CSF RHINORHEA

VS

SINUSITIS

CAUSE

TYPE OF D/S

SNIFFING

GLUCOSE

MALIGNANCY NOSE

M/C BENIGN TUMOUR - CAPILLARY HEMANGIOMA (SITE-LITTLE AREA)

M/C SITE CAVERNOUS HEMANGIOMA -IT

M/C SITE MALIGNANT MELANOMA- NASAL SEPTUM

M/C SITE PAPILLOMA - SKIN BEHIND COLUMELLA/VESTIBULE

M/C MALIGNANT -SCC —> LATERAL WALL OF NOSE

ALL TUMOUR TREATMENT —> SURGERY USING LATERAL RHINOTOMY/MOORE'S INCISION



INVERTED PAPILLOMA/RINGERTZ TUMOUR/SCHNIDERIAN PAPILLOMA/TRANSITIONAL CELL PAPILLOMA

TUMOUR THAT ARISES FROM LATERAL WALL OF NOSE

PREMALIGNANT TUMOUR

M>F

>40 YEARS

INVERTED AS PER HPE FINDING

IN BOTH SCC AND INVERTED PAPILLOMA THERE IS INVASION OF MAXILLARY SINUS

BUT IF

WALL OF MAX SINUS DAMAGE PRESENT - SCC

IF NOT-I P

M/C BENIGN TUMOUR OF PNS-OSTEOMA- SEEN IN
FRONTAL SINUS

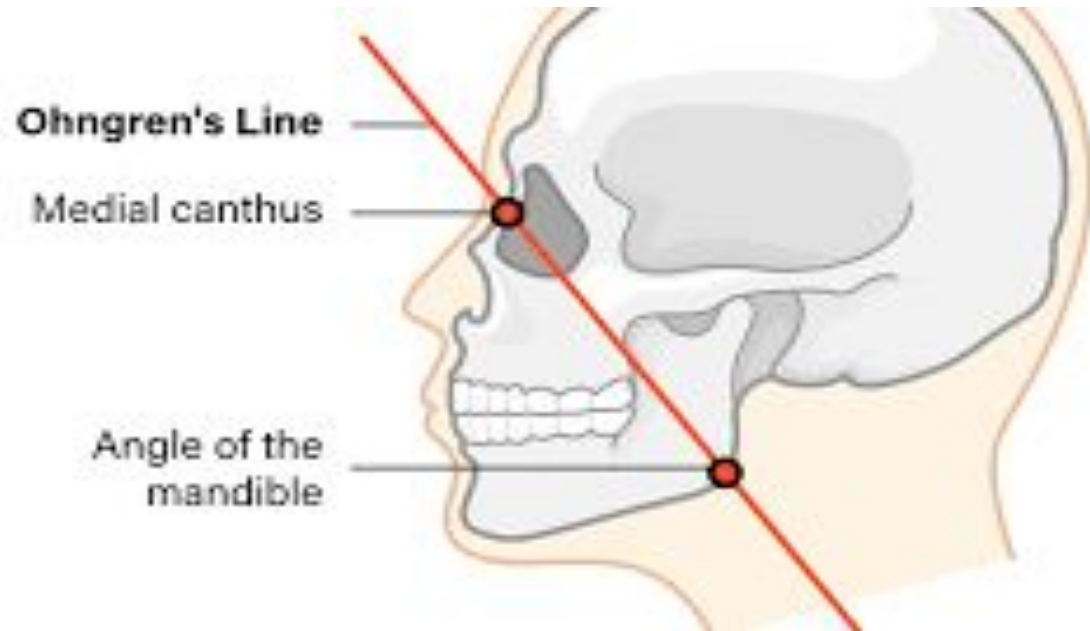
M/C MALIGNANT TUMOUR OF PNS-SCC - IN
MAXILLARY SINUS - PEOPLE WORKING IN NICKEL
INDUSTRY

2ND M/C IS ADENO CARCINOMA - IN ETHMOID
SINUS- WOOD INDUSTRY

FOR THESE T/T IS SURGERY + RADIOTHERAPY

MAXILLARY SINUS CA

OGHRENS CLASSIFICATION



GRANULOMATOUS DISEASES OF NOSE

M/C SITE OF LEPROSY AND TB - ANT.PART OF SEPTUM AND I.T

STAGE OF SYPHILLIS INVOLVING NOSE - TERTIARY

M/C SYMPTOM - NASAL OBSTRUCTION

LUPUS VULGARIS-M/C SITE IS MUCOCUTANEOUS JUNCTION OF NASAL SEPTUM

RHINOPHYMA



RHINOSCLEROMA(Frisch bacillus)

K . RHINOSCLEROMATIS

WOODY NOSE/TAPIR NOSE/HEBRA NOSE/TAPIR NOSE

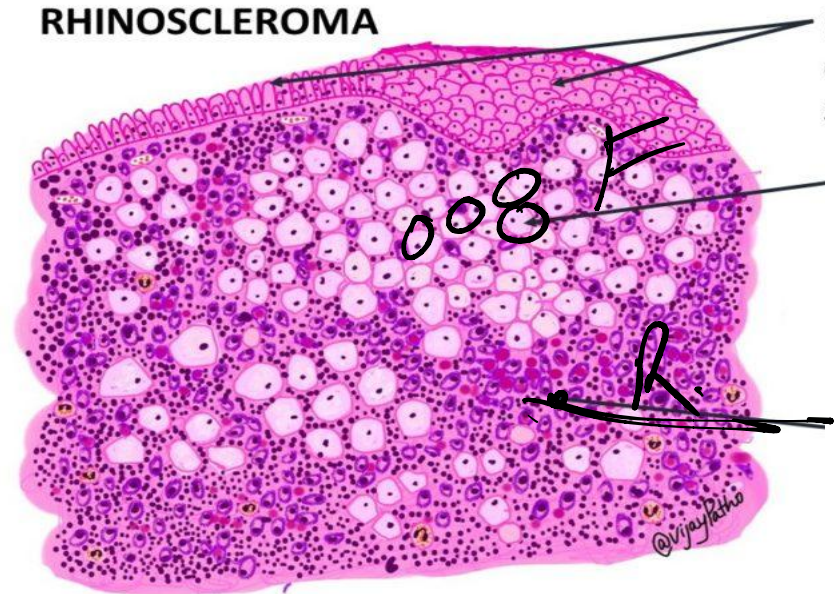
STAGES

ATROPHIC- CARPENTER GLUE LIKE
DISCHARGE

GRANULOMATOUS- HEBRA NOSE

CICATRICAL- TAPIR NOSE

RUSELL BODIES AND MIKULICZ CELLS



RHINOSCLEROMA A/W SUBGLOTTIC STENOSIS WITH
RESPIRATORY FAILURE

T/T OR DOC

TETRACYCLINE + STREPTOMYCIN

Hebra nose



RHINOSPORODIOSIS

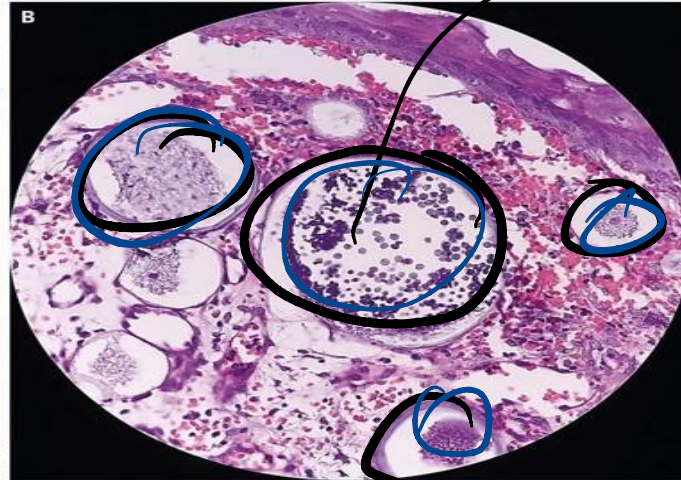
SOUTH INDIA, SEA

EXCISION AND CAUTERY AT BASE

DAPSONE IS THE DOC

MULBERRY LIKE POLYP

Spherules.



NASAL POLYP

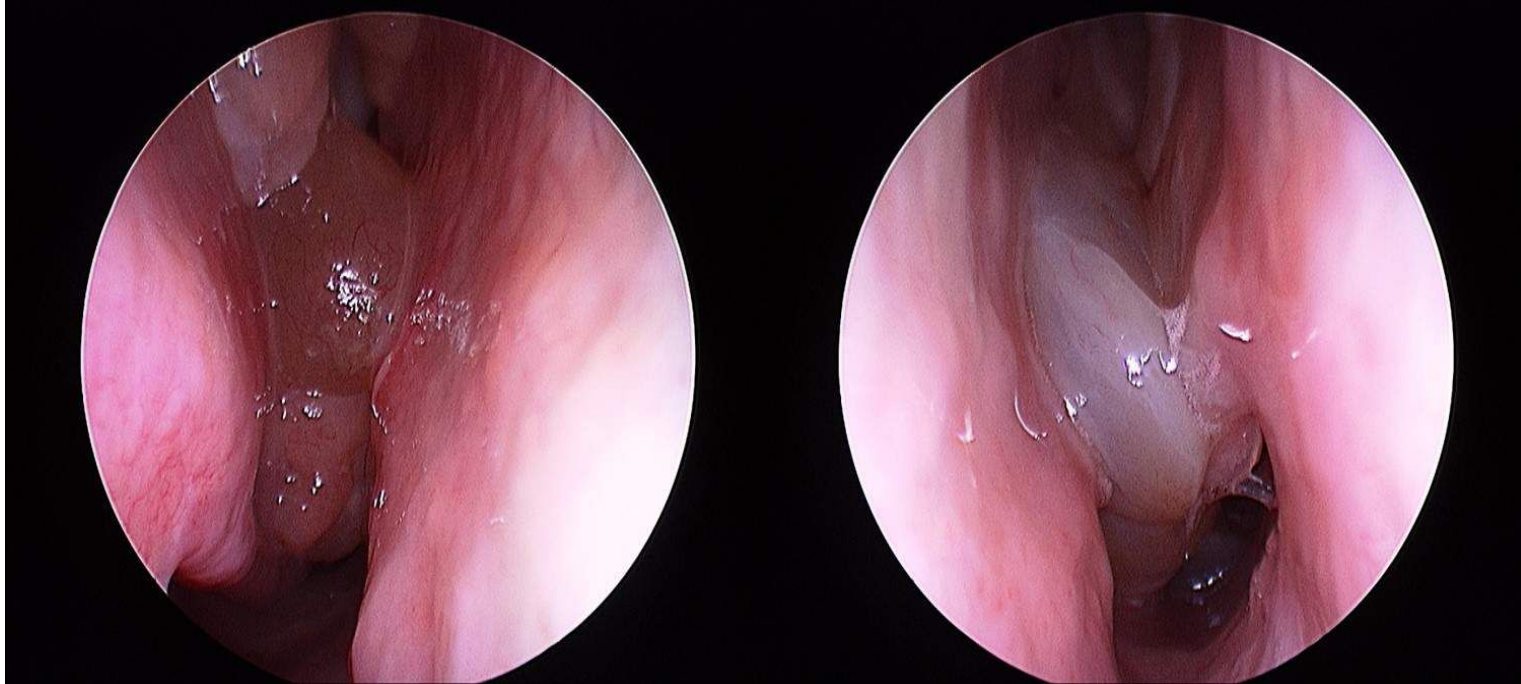
BERNOULIS PRINCIPLE

KARTAGENER SYNDROME- NASAL POLYP + BRONCHIECTASIS +
DEXTROCARDIA

YOUNG SYNDROME- NP + AZOOSPERMIA + BRONCHIECTASIS

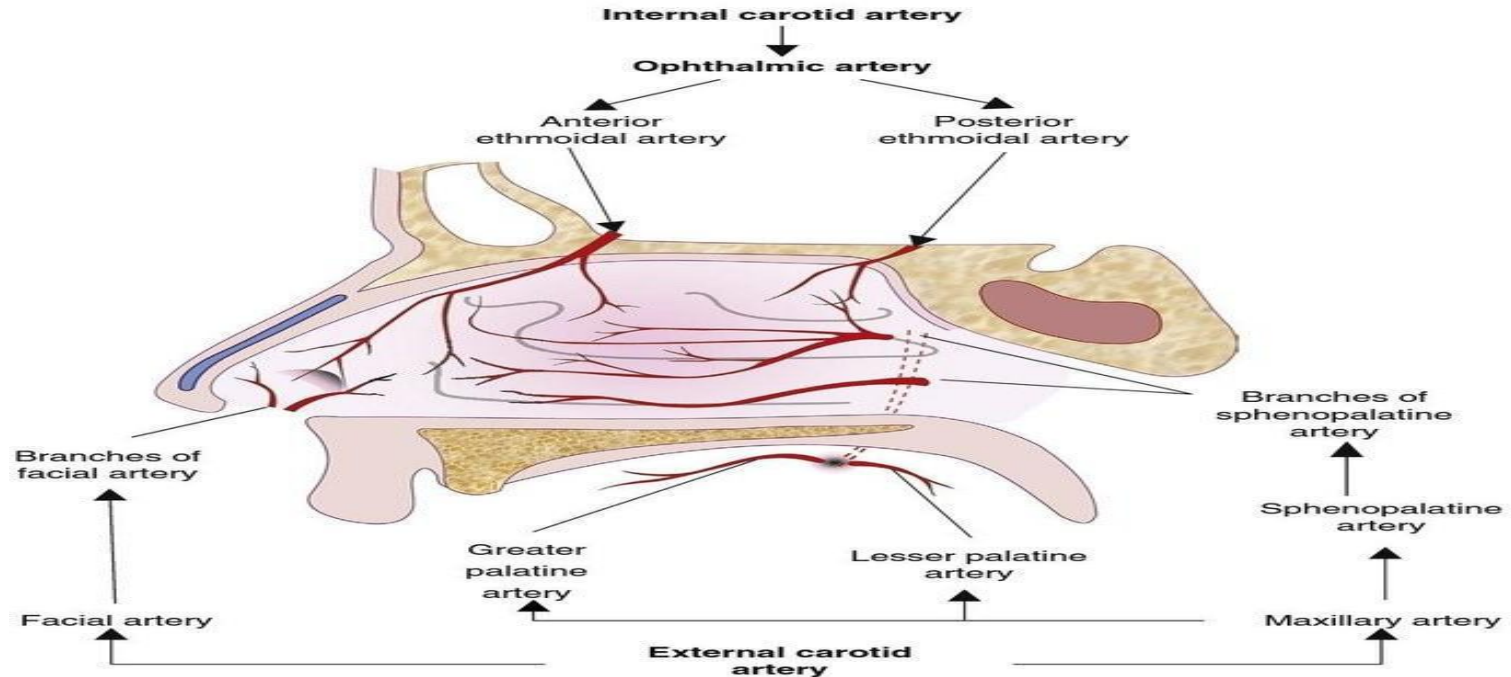
SAMTERS TRIAD - NP + BRONCHIAL ASTHMA+ ASPIRIN SENSITIVITY

DIFFERENCE BETWEEN ANTROCHOANAL AND ETHMOIDAL POLYP



BLOOD SUPPLY OF NASAL SEPTUM & EPISTAXIS

ANT ETHMOIDAL ARTERY EMBOLISATION DONE IF BLEEDING IS FROM ABOVE MIDDLE TURBINATE



Rx:

1. Pinch the nose for 3-5 minute (then)
2. Chemical cauterization of little's area with AgNO_3 . (then)
 - *AgNO_3 is cauterization agent it will burn that area and bleeding will stop.*
3. Anterior nasal packing (on both side) (if this failed)
4. Posterior + anterior nasal packing.
5. If the nasal packing failed then we will go for. 5. Endoscopic ligation of Sphenopalatine artery (SP) Q (1st artery to ligation)
 - If SP failed then we will go for maxillary artery ligation – if its failed – external carotid artery ligation - if its failed – ethmoidal artery ligation.

SEPTAL HAEMATOMA/SEPTAL ABSCESS

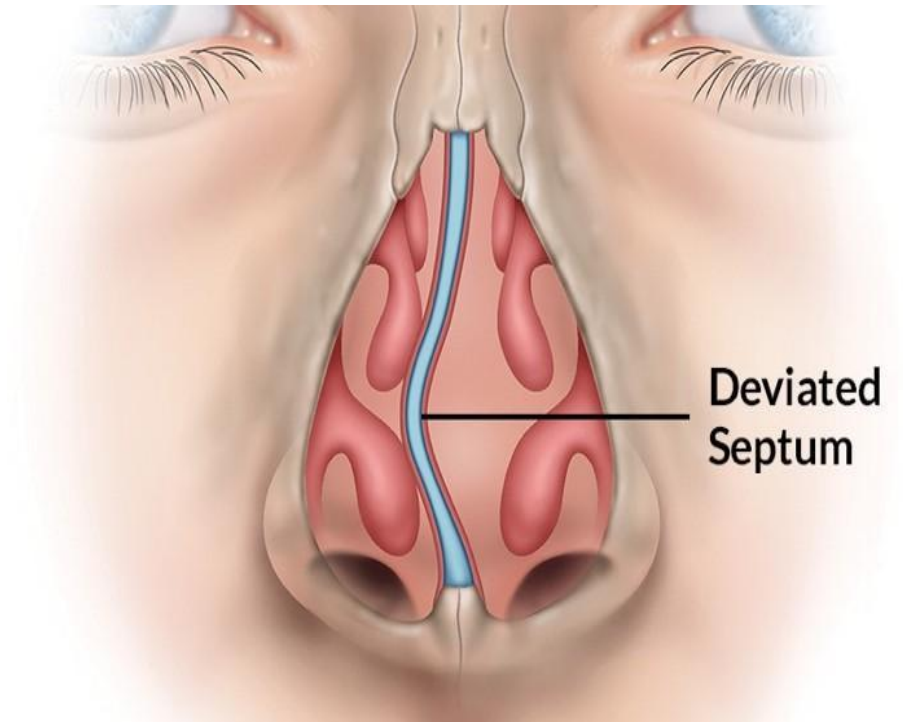


DNS

COMPENSATORY HYPERTROPHY

SLUDDERS NEURALGIA

SEPTOPLASTY V/S SMR



RHINOPLASTY

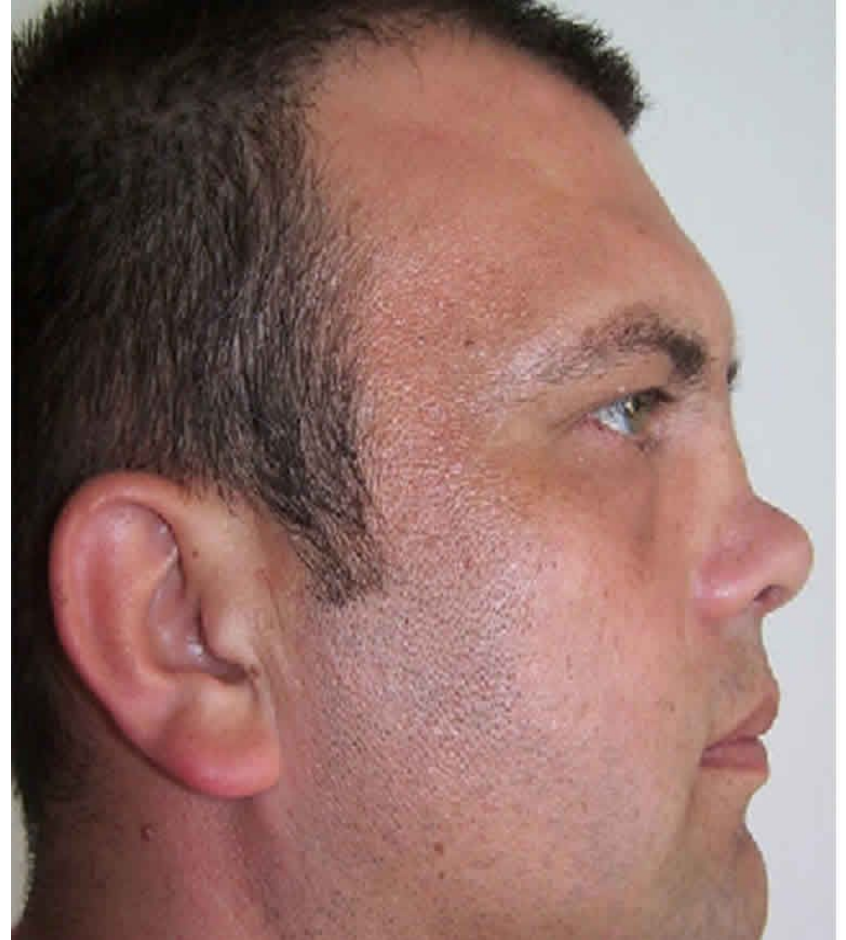
DEPRESSION OF DORSUM-
SADDLE NOSE

CAUSE - SURGICAL TRAUMA

TREATMENT-

AUGMENTATION RHINOPLASTY

- FROM ILIAC CREST



SWELLING OF DORSUM - HUMP NOSE

MAINLY IS CONGENITAL

DP REDUCTION RHINOPLASTY

M/C COMPLICATION OF ENDOSCOPIC SINUS SURGERY IS

SYNECHIAE

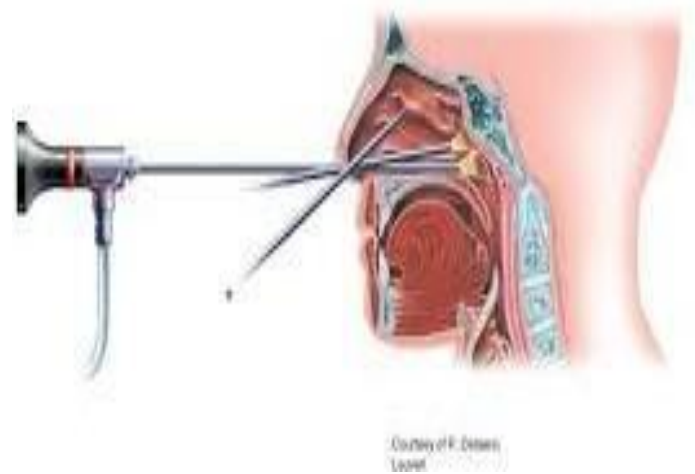
TREATMENT - MITOMYCIN C

ENDOSCOPE

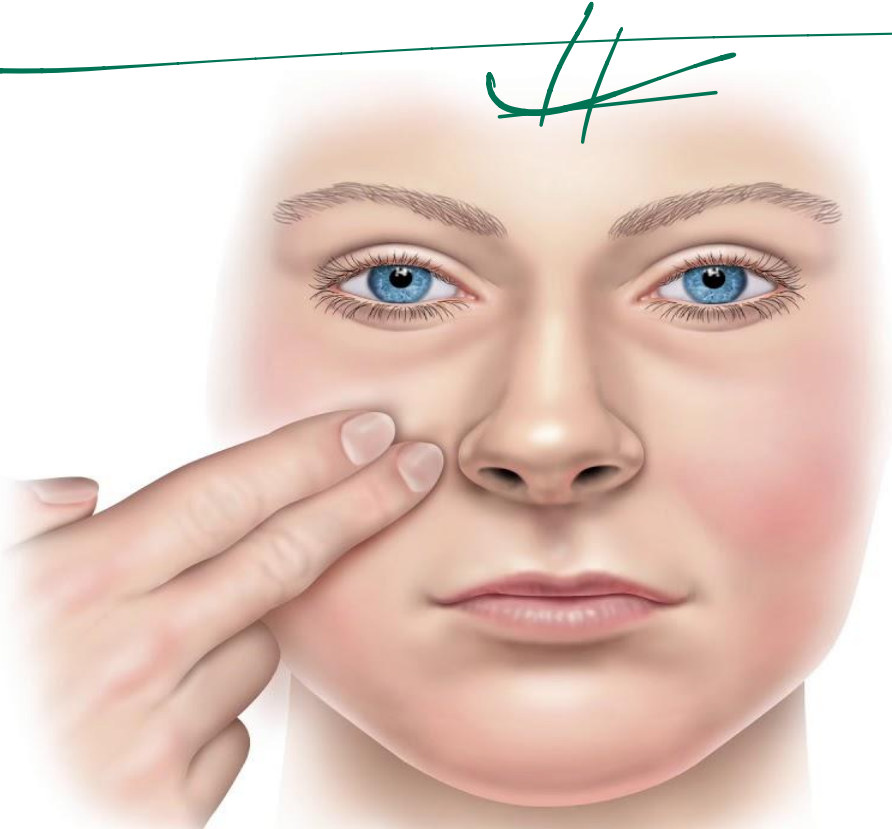
DIAMETER OF ENDOSCOPE- 4MM , 2.7 MM

A—>P STAMBERGER APPROACH

P—> A WIGAND APPROACH



COTTLE TEST-NASAL VALVE PATENCY TEST



RHINOLALIA CLAUSA - HYPONASALITY →
DECREASED NASAL COMPONENT → NASAL
OBSTRUCTION

⇒ Voice when Nose
is
Blocked.

RHINOLALIA APERTA → HYPERNASALITY →
INCREASED NASAL COMPONENT → PALATE
INSUFFICIENCY

Nose and
Mouth

eg: Cleft palate.

XRAY- PNS

WATERS VIEW / O.M VIEW

BEST FOR MAXILLARY SINUS

PIERES VIEW(W+OPEN MOUTH)

FOR SPHENOID SINUS

CALDWELL VIEW

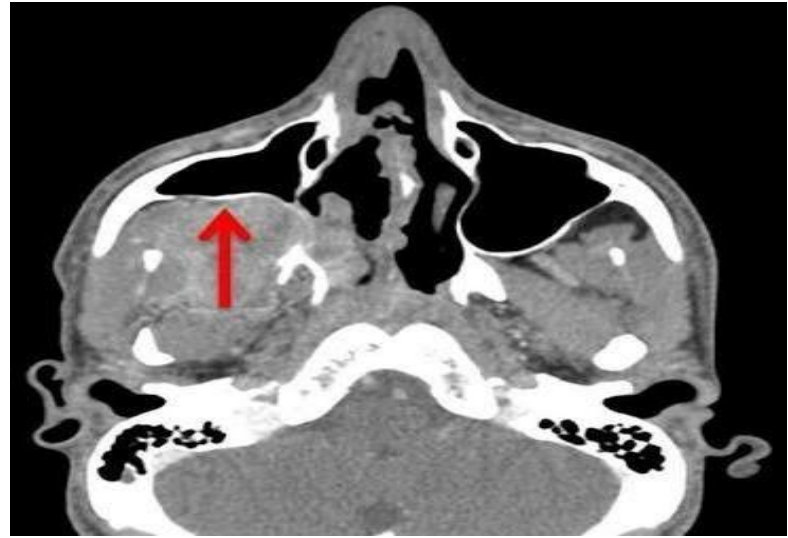
FOR FRONTAL AND ETHMOID SINUS

JNA

SPHENOPALATINE FORAMEN

HOLLMAN MILLER SIGN

DUMBELL SHAPED TUMOUR-



PTERYGOPALATINE FOSSA → INFRATEMPORAL FOSSA

PRE OPERATIVE EMBOLISATION IS DONE TO KNOW WHICH IS THE BLOOD VESSEL SUPPLYING THIS TUMOUR - MAXILLARY ARTERY

HOLLMAN MILLER SIGN - ANT. BOWING OF THE POST WALL OF MAXILLARY ANTRUM

NASOPHARYNGEAL CARCINOMA

RADIOSENSITIVE

ASSOCIATED ETIOLOGY WITH EBSTEIN BARR VIRUS

FOSSA OF ROSENMUELLER

TROTTERS TRIAD(NPC)- NEURALGIA PALATAL PALSY CONDUCTIVE HL

SEPTAL PERFORATION

CAUSES:

TRAUMA(M/C)

SURGERY

TB

LEPROSY



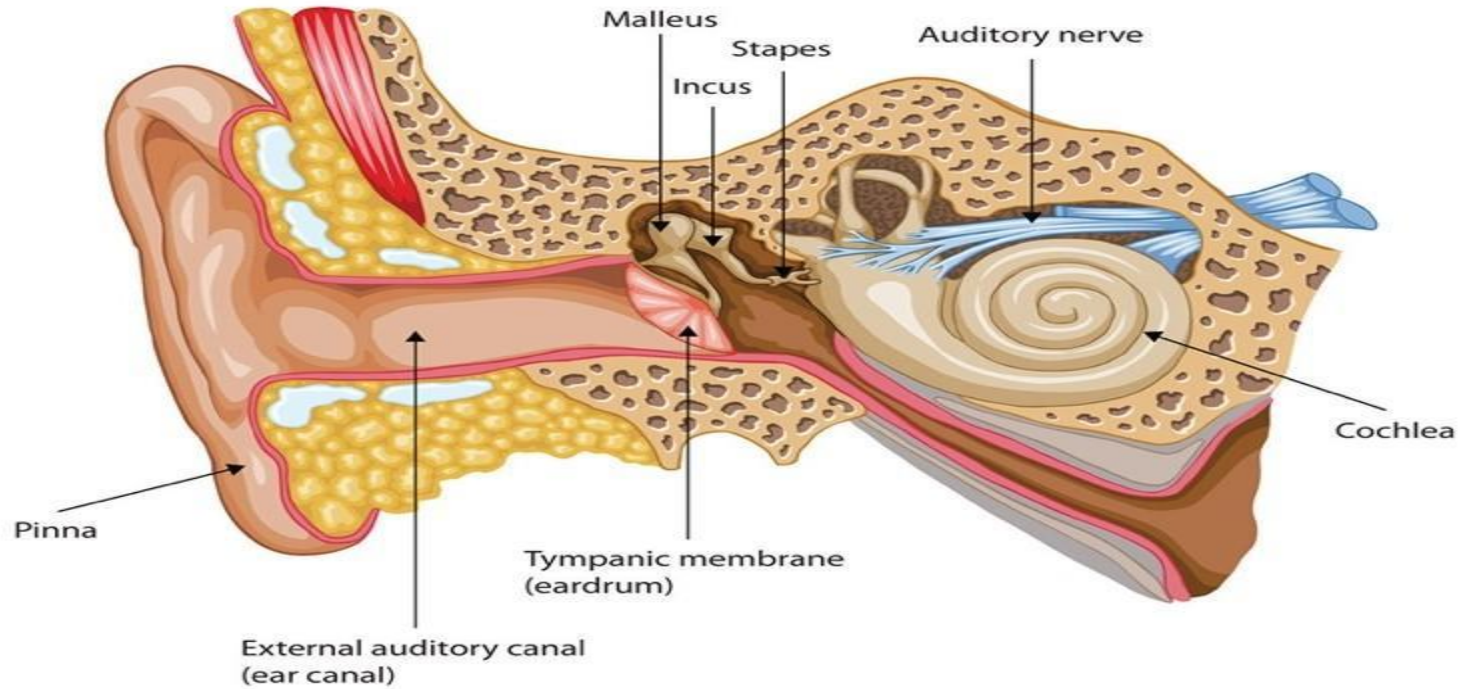
TB,LEPROSY,LUPUS CAUSES PERFORATION IN CARTILAGE PART

TERTIARY SYPHILIS CAUSES BONY PERFORATION

UPSIT- UNIVERSITY OF PENNSYLVANIA SMELL IDENTIFICATION TEST

- BEST TEST FOR OLFACTION

EAR



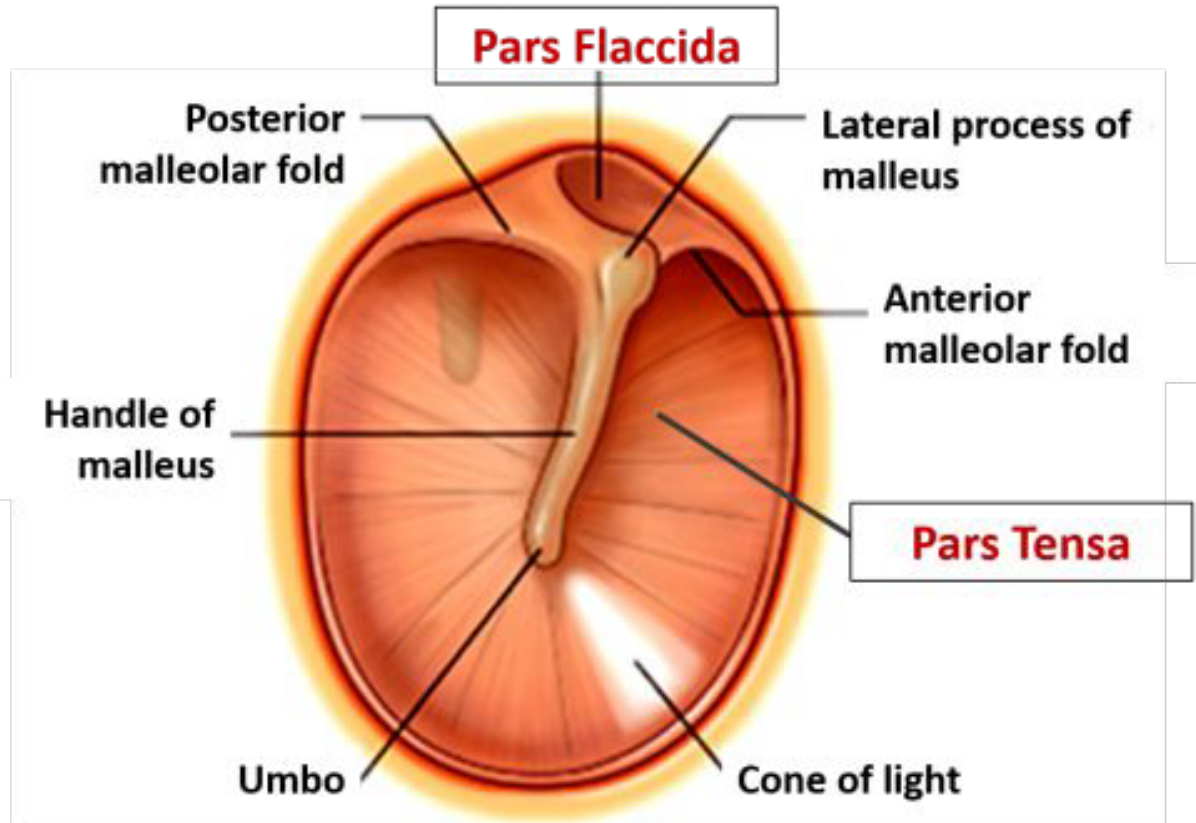
BOUNDARIES OF MIDDLE EAR

1ST BRANCHIAL ARCH (ECTODERM)--> EXT. EAR

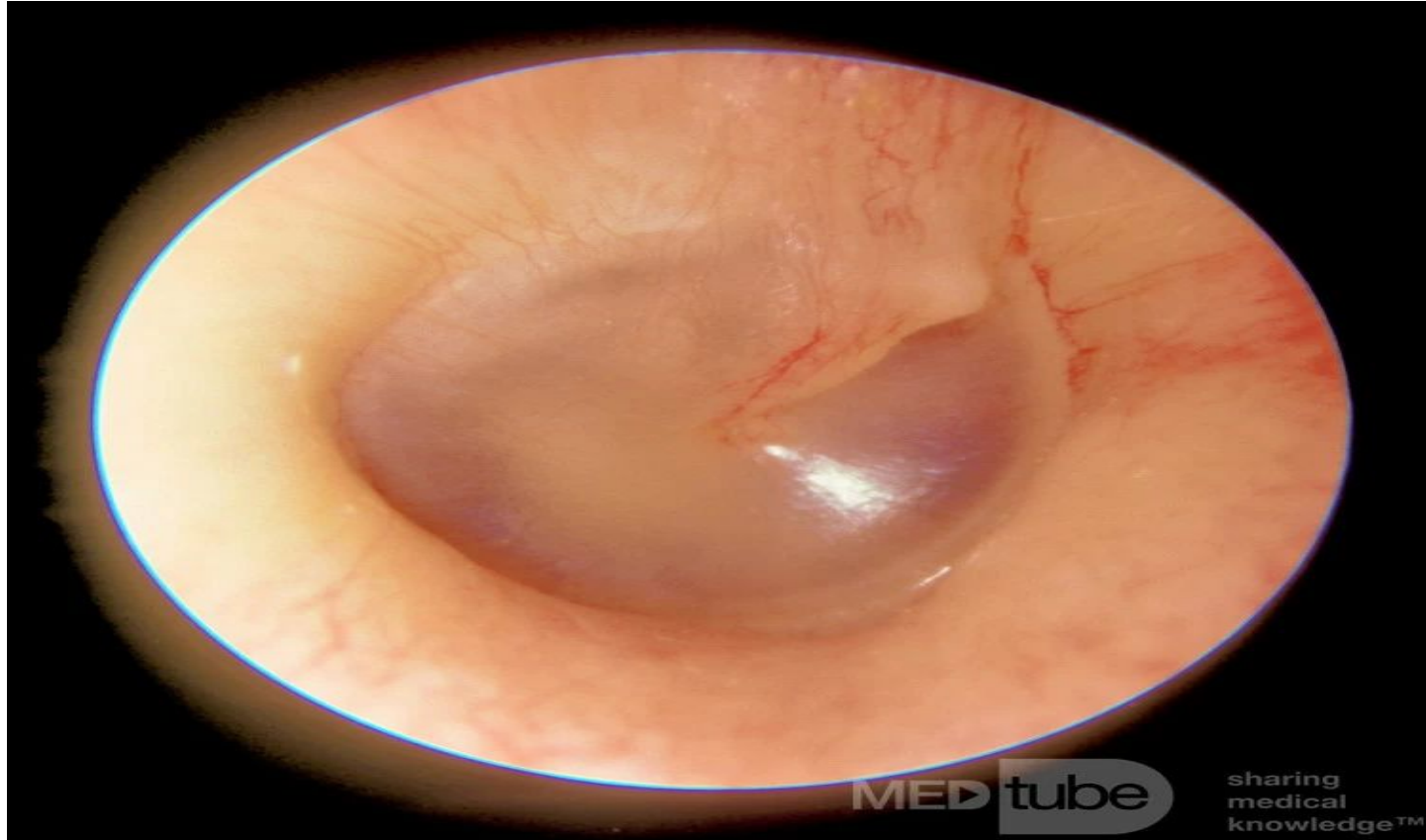
2ND BRANCHIAL ARCH(ENDODERM)--> MIDDLE EAR

OTIC CAPSULE(NEUROECTODERM)--> INNER EAR

TYMPANIC MEMBRANE AND ITS PARTS



CONE OF LIGHT CLOCK POSITION



T M MARKERS

CONCAVE IN SHAPE

UMBO(BEST MARKER)

CONE OF LIGHT -AI QUADRANT

HANDLE OF MALLEUS

ARTERIAL SUPPLY

MAXILLARY ARTERY

NERVE SUPPLY OF TM

OUTER SURFACE

ANT HALF - AURICULOTEMPORAL NERVE

POST HALF- AURICULAR BRANCH OF VAGUS

INNER SURFACE

TYMPANIC BRANCH OF GLOSSOPHARYNGEAL NR.

OVAL AND ROUND WINDOW AND THIRD WINDOW
EFFECT

SEEN IN SCCD-SEMI CIRCULAR CANAL DEHISCENCE

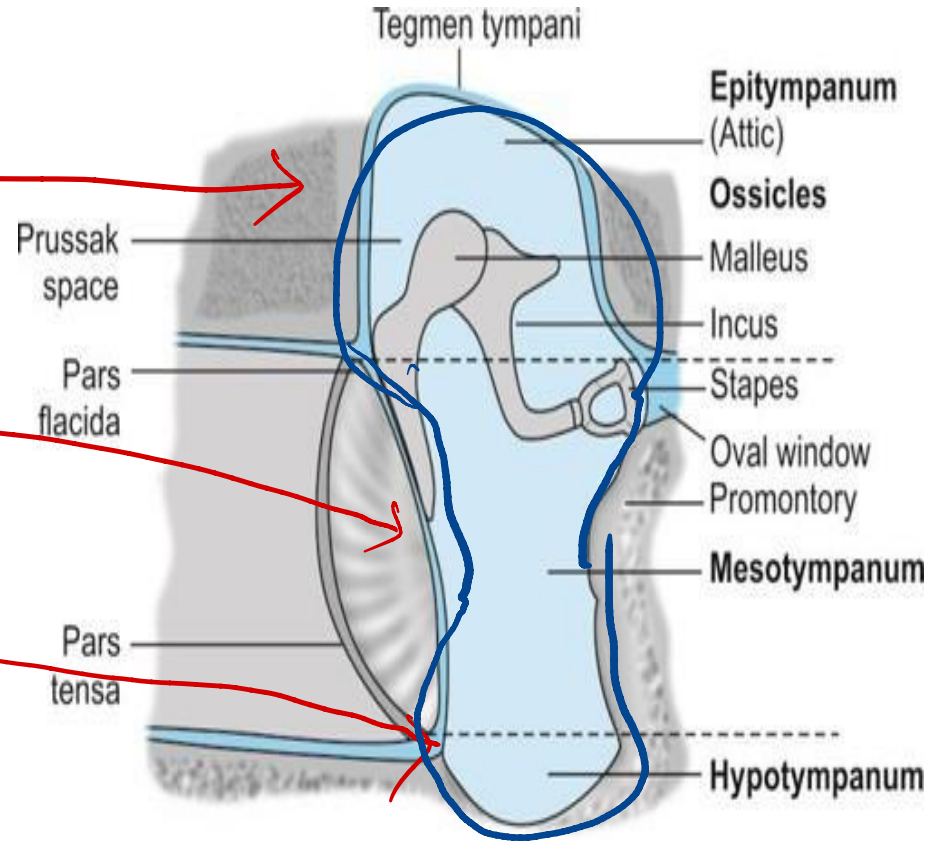
PROTYMPANUM

PART OF MIDDLE EAR SURROUNDING EUSTACHIAN TUBE

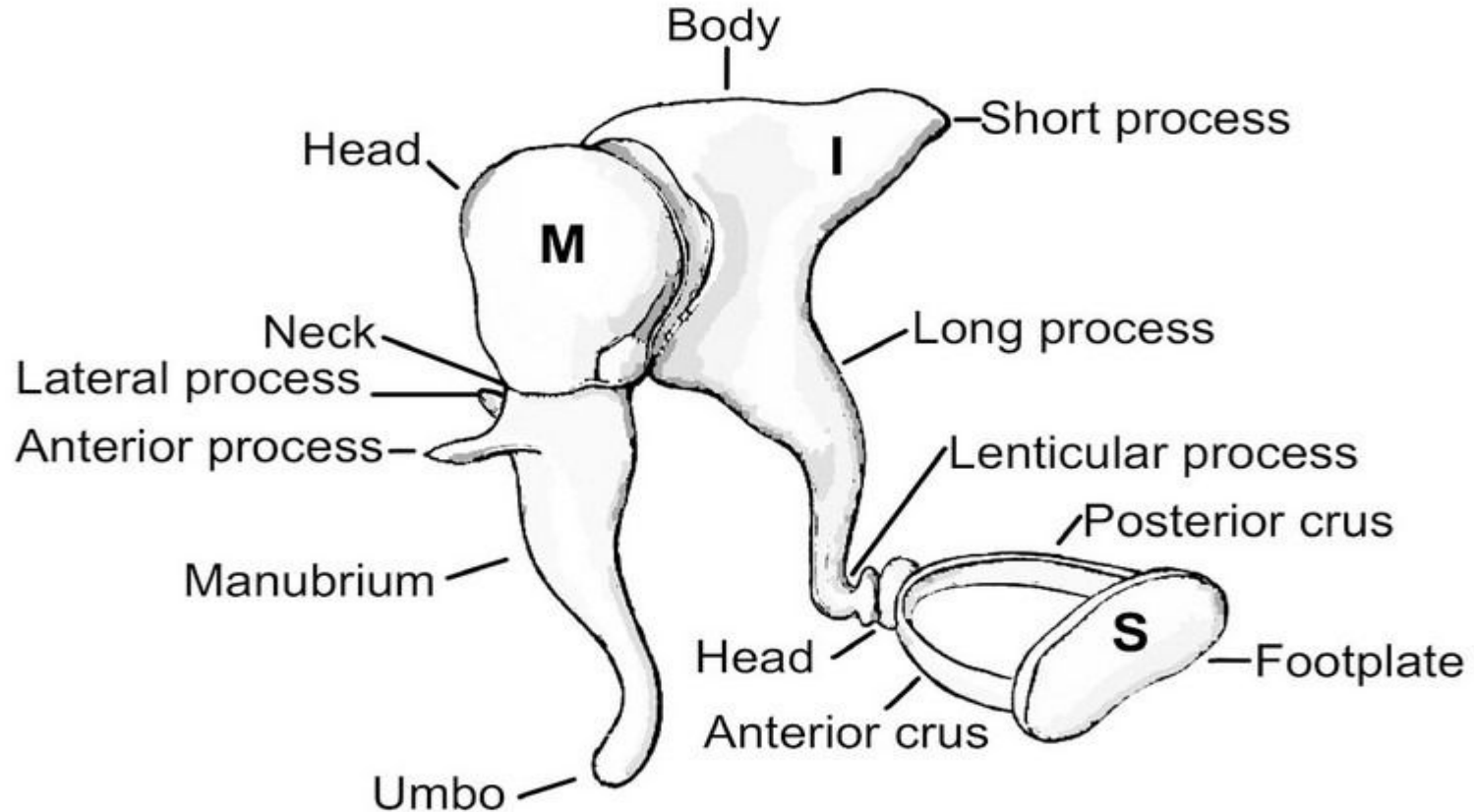
EPITYMPANUM-6MM

MESOTYMPANUM-2MM

HYPOTYMPANUM-4MM



OSSICLES AND ITS PARTS



LONG PROCESS OF INCUS AKA LENTICULAR PROCESS

LENTICULAR PROCESS IS M/C SITE OF NECROSIS

IS JOINT IS BALL AND SOCKET JOINT → MOST PRONE TO TRAUMA

FOOT PLATE OF STAPES → M/C SITE OF OTOSCLEROSIS

MALLEUS AND INCUS DEVELOP FROM 1ST BRANCHIAL ARCH

STAPES FROM 2ND BRANCHIAL ARCH

EXCEPT MEDIAL SURFACE/FOOT PLATE THAT DEVELOPS FROM OTIC CAPSULE

NERVE SUPPLY OF MIDDLE EAR

SENSORY

9TH NR. JACOBSON

10TH NERVE- ARNOLD

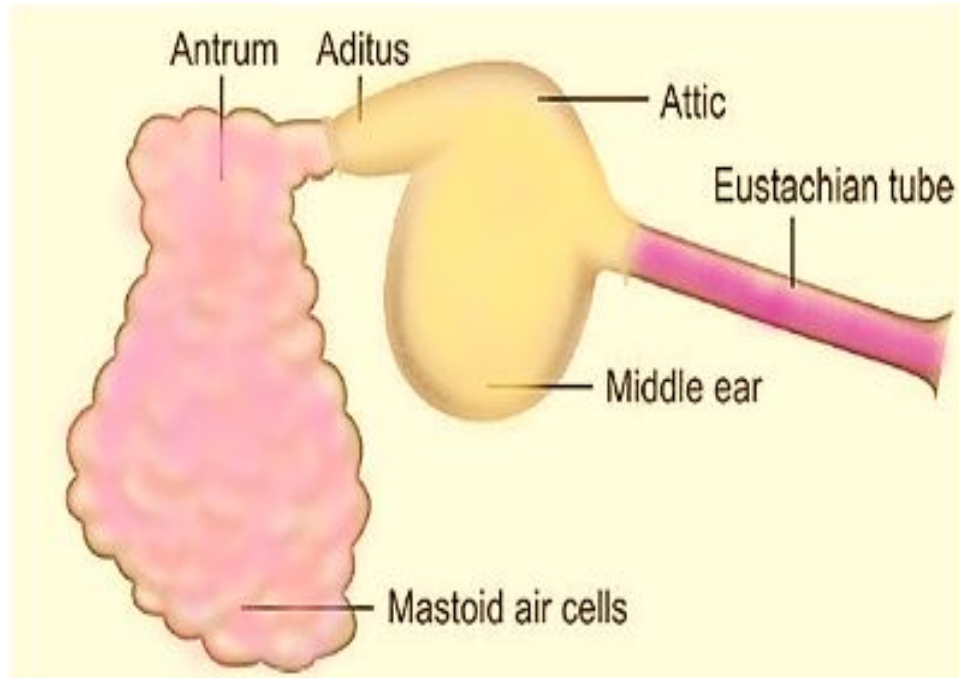
TOGETHER FORM TYMPANIC PLEXUS

MOTOR

TENSOR TYMPANI SUPPLIED BY MANDIBULAR DIVISION OF TRIGEMINAL NR.

STAPEDIUS BY 7TH NR.

WHAT IS MIDDLE EAR CLEFT



THICKNESS OF MASTOID IN ADULT IS 15 MM

THICKNESS OF MASTOID IN NEWBORN IS 1-2 MM

GROWS BY THE RATE OF 1-2 MM PER YEAR SO IN 15 YEARS REACHES
ADULT THICKNESS

TIP IS ABSENT IN NEWBORN WHERE FACIAL NR. IS RUNNING SO IT IS
EXPOSED

KORNER SEPTUM

PETROSQUAMOUS SUTURE LINE

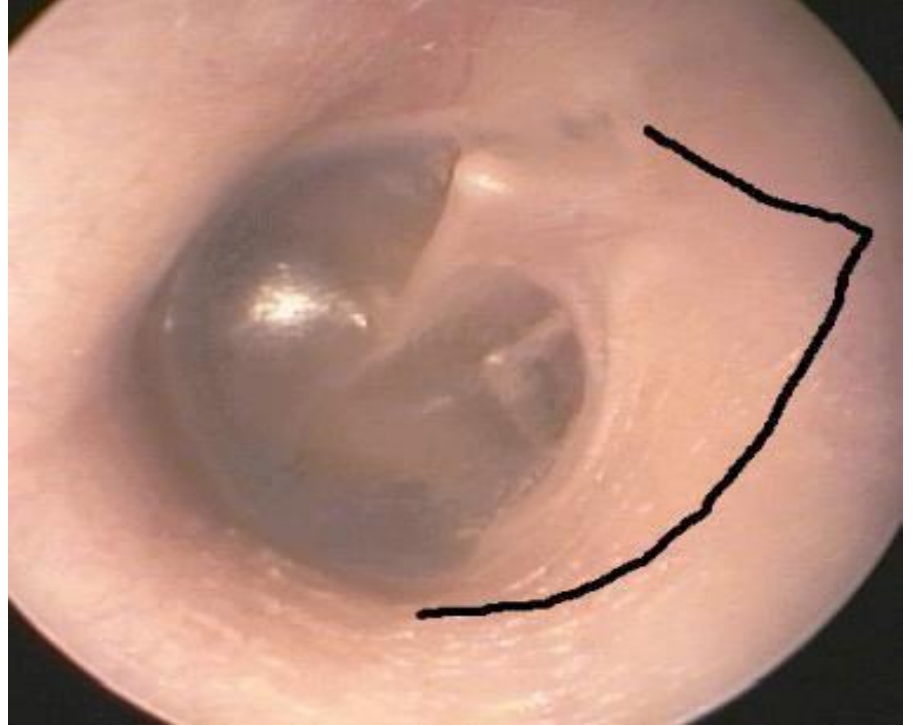
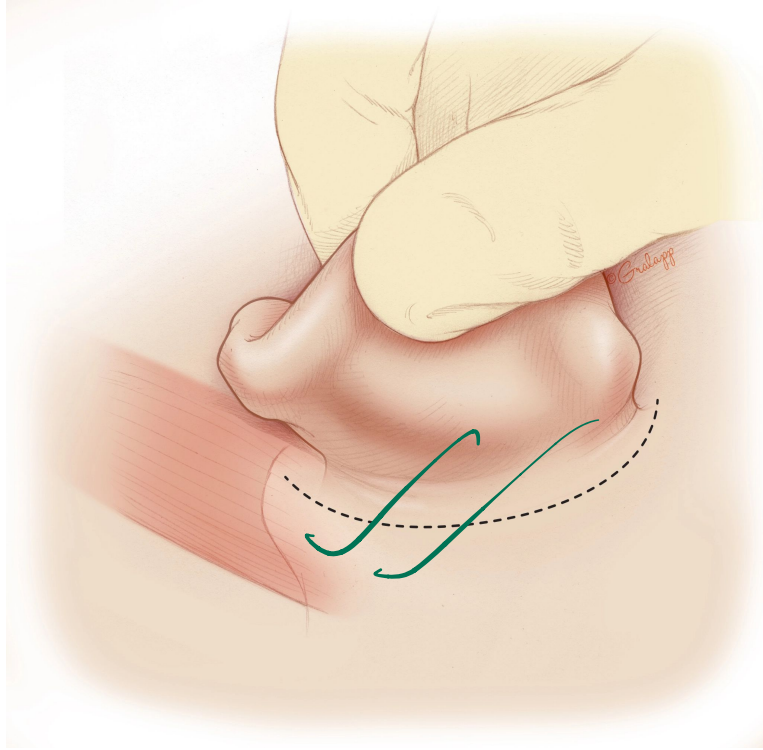
TO REACH MIDDLE EAR

MASTOID

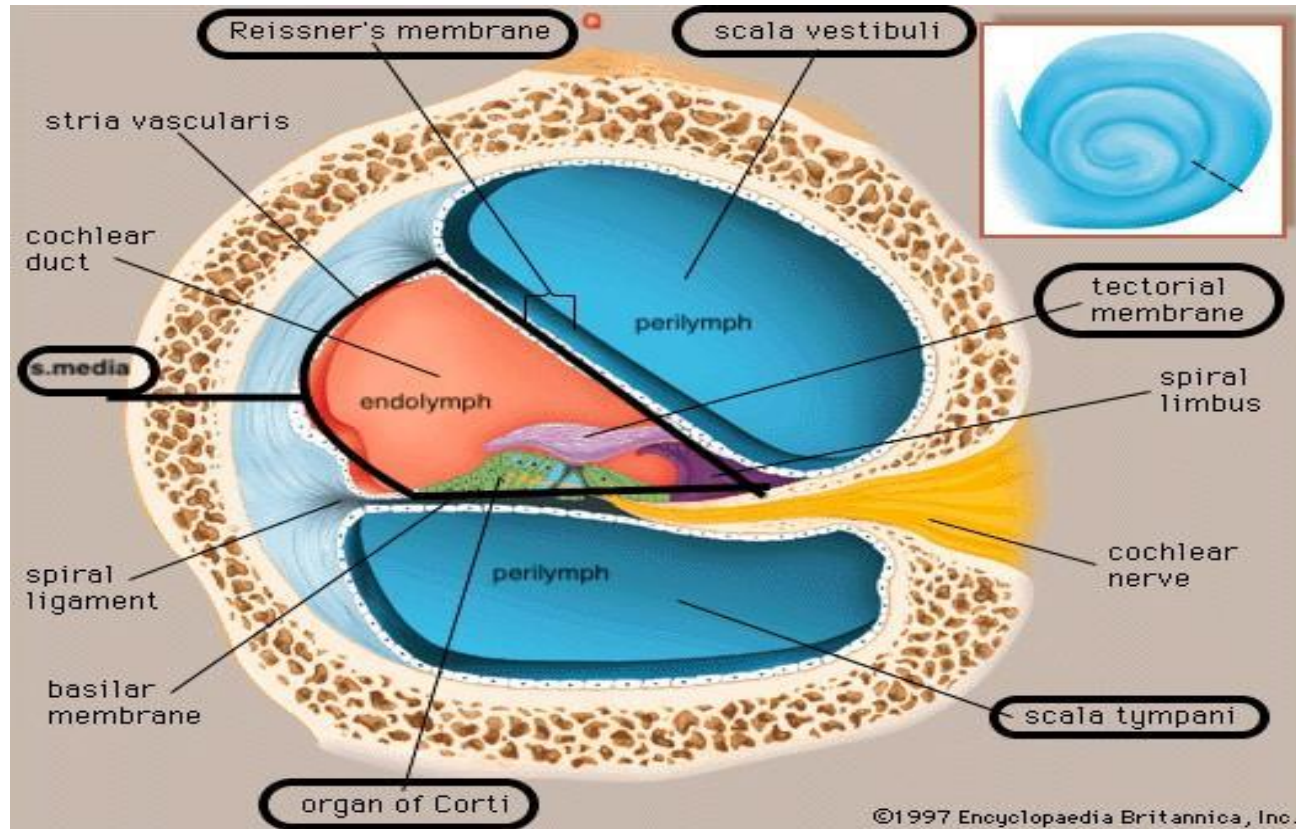
(POST AURICULAR APPROACH)-WILDES INCISION

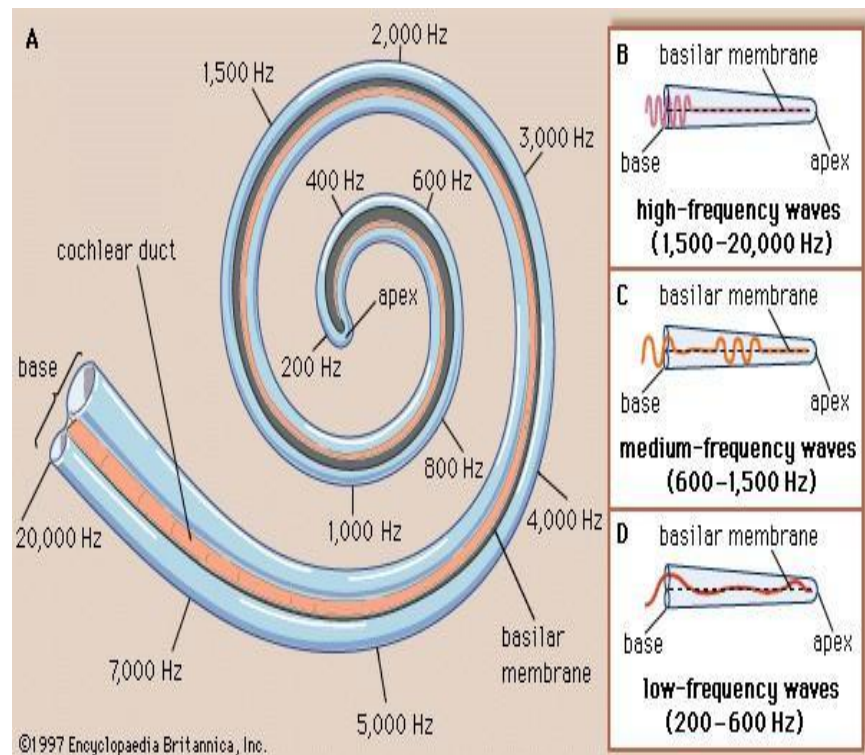
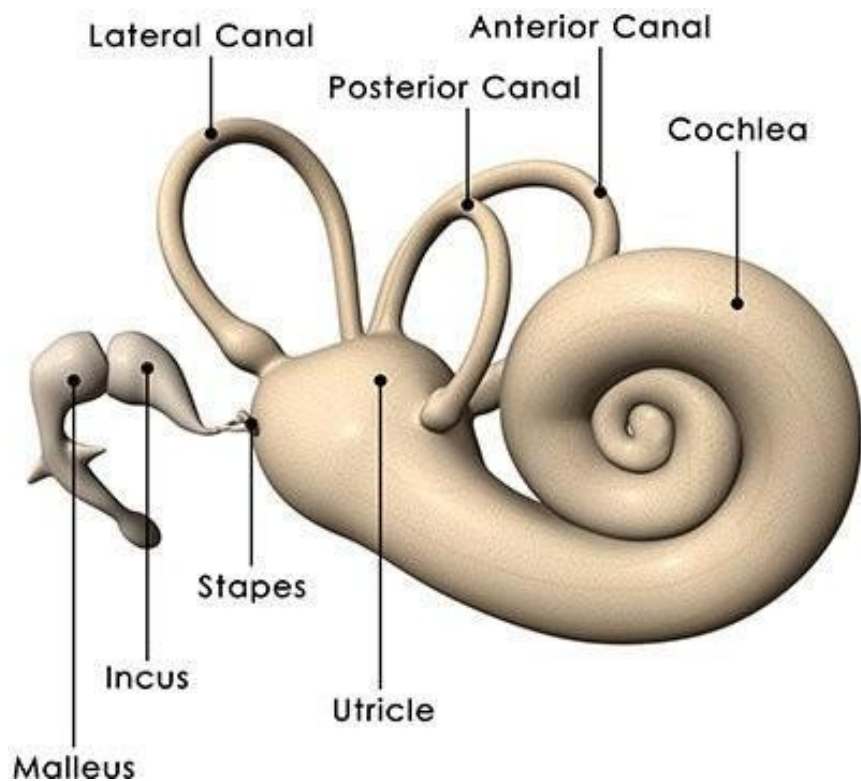
ANNULUS

(ENDOMEATAL APPROACH)-ROSEN INCISION



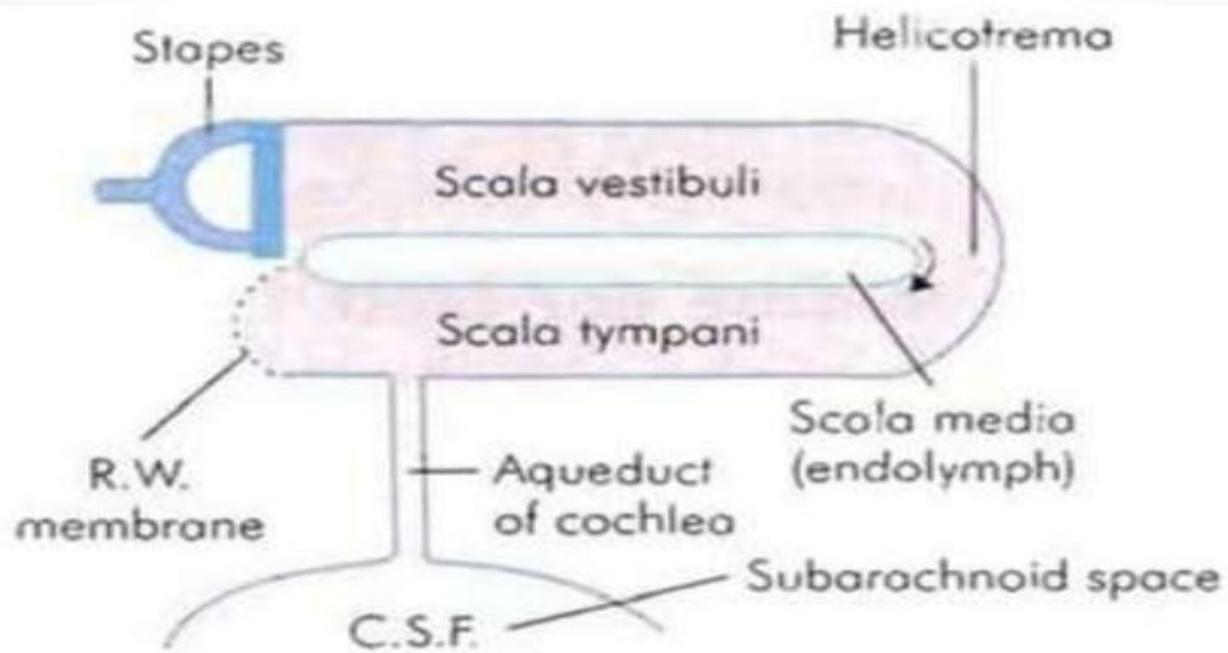
INNER EAR (LABYRINTH)

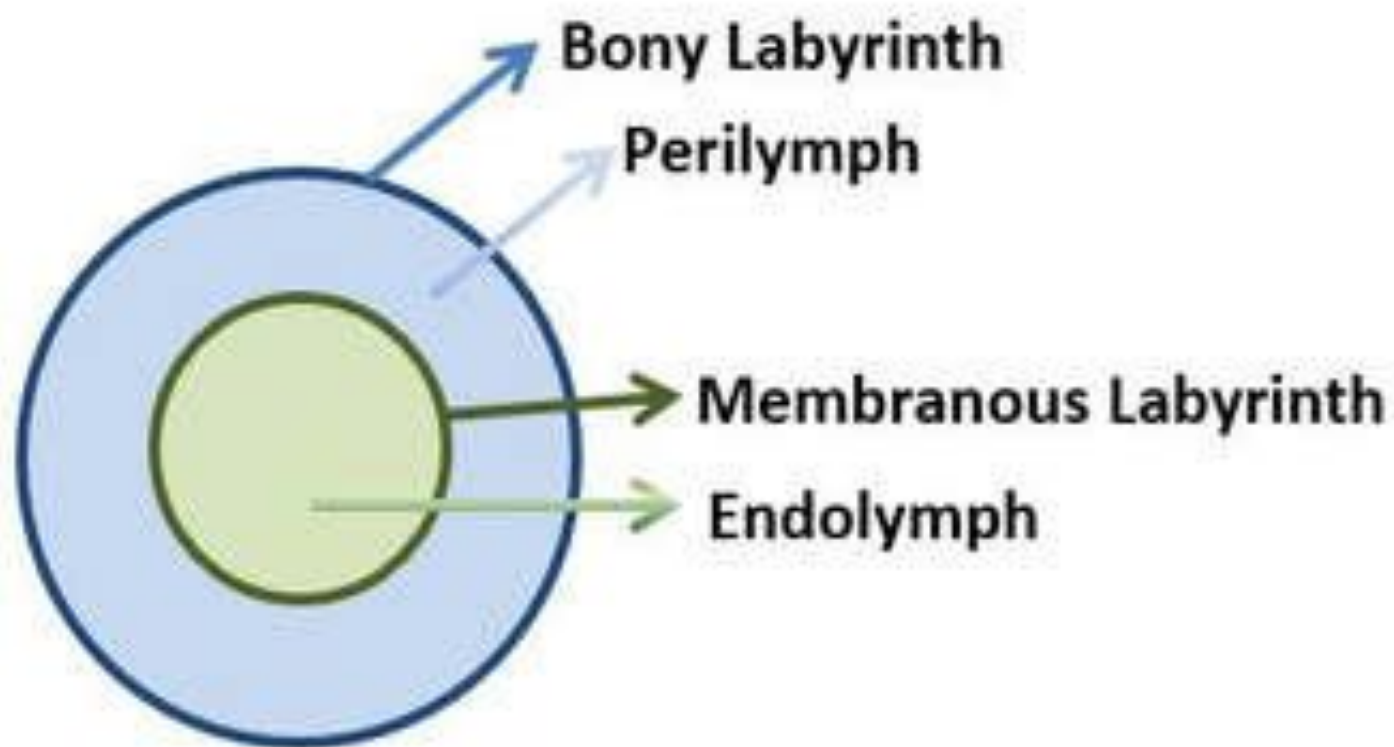




Semi Circular Canals

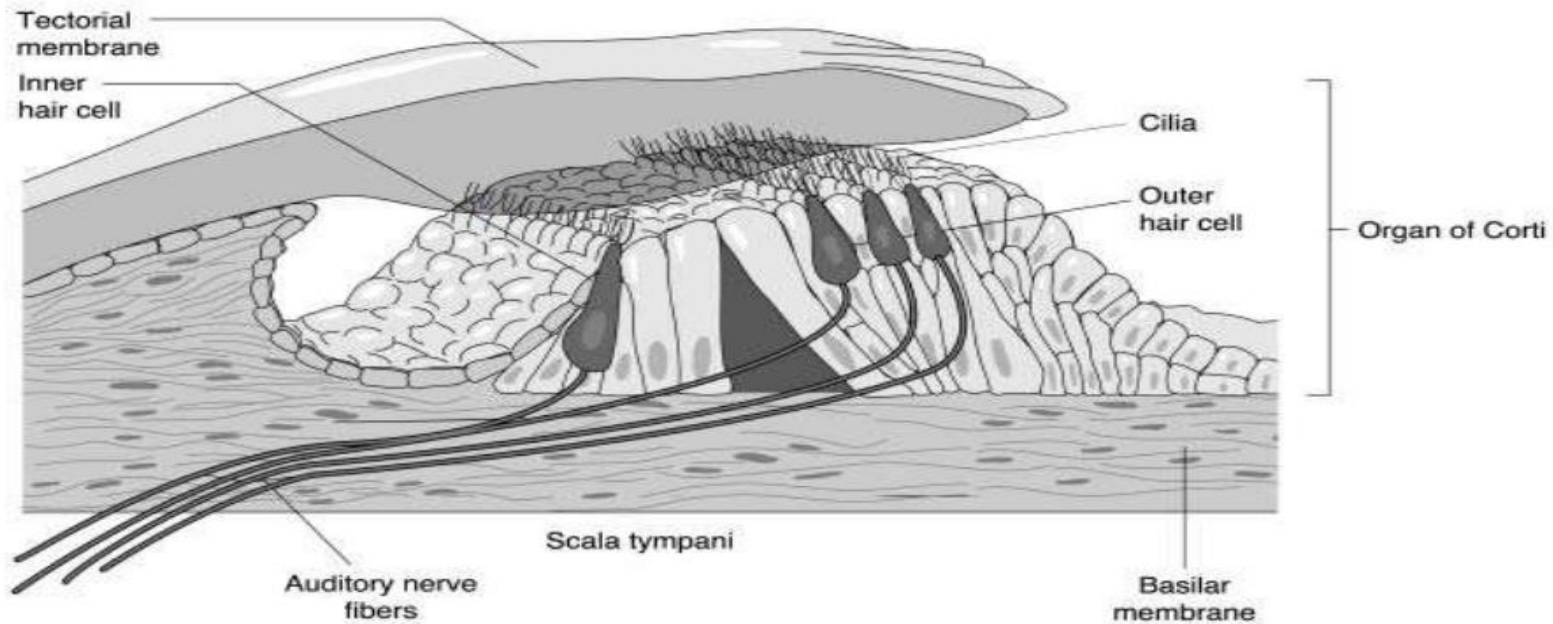
- 3 semi circular canals
 - Superior
 - Posterior
 - Lateral
- They lie in planes at right angle to each other.
- Each canal got ampullary end ,which open in the vestibule.
- Non-ampullary end of Lateral SCC open independently
- Post. SCC and Sup. SCC form a common opening Called *CRUS COMMUNE*





ORGAN OF CORTI

- Organ of hearing,
- Situated on basilar membrane of cochlea
- 3 parts
 - Sensory Hair cells
 - Supporting cells
 - Tectorial Membrane



Inner Hair cells

- Total No : 3500
- Arranged in single row
- Flask shaped
- 95% afferent from Cochlear nerve (8th CN)
- Stereocilia arranged in V or W format
- More resistant
- Transmit auditory stimuli

Outer Hair cells

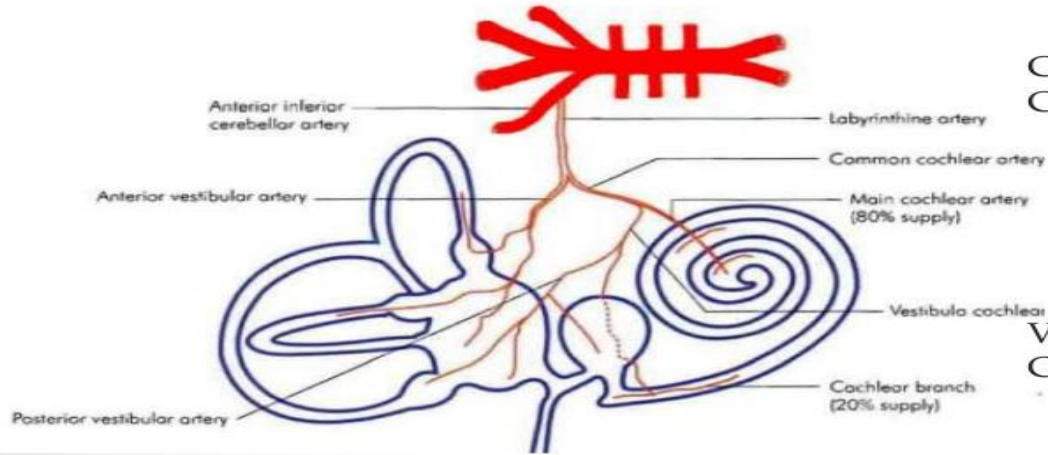
- 12,000 [1:4 ratio]
- Multiple rows
- Cylindrical
- Efferent from Sup. Olivary complex
- Stereo cilia single continuous layer.
- Easily damaged to drugs & loud sound
- Modulate the IHC.

Supporting cells

- Stabilize and support the organ of corti
- Supply nutrient to the highly specialized sensory cells(inner and outer hair cells)
- Includes
 - Hensen's Cell
 - Claudius Cells
 - Deiters Cells
 - Boettcher's Cell

Blood Supply

Arterial



Ant. Inferior Cerebellar
Art



Labyrinthine/Intern
al Auditory art.

Common
Cochlear

Ant.
Vestibular
A
(to utricle &
LSCC, SSCC)

Vestibulo
Cochlear

Main Cochlear
80%

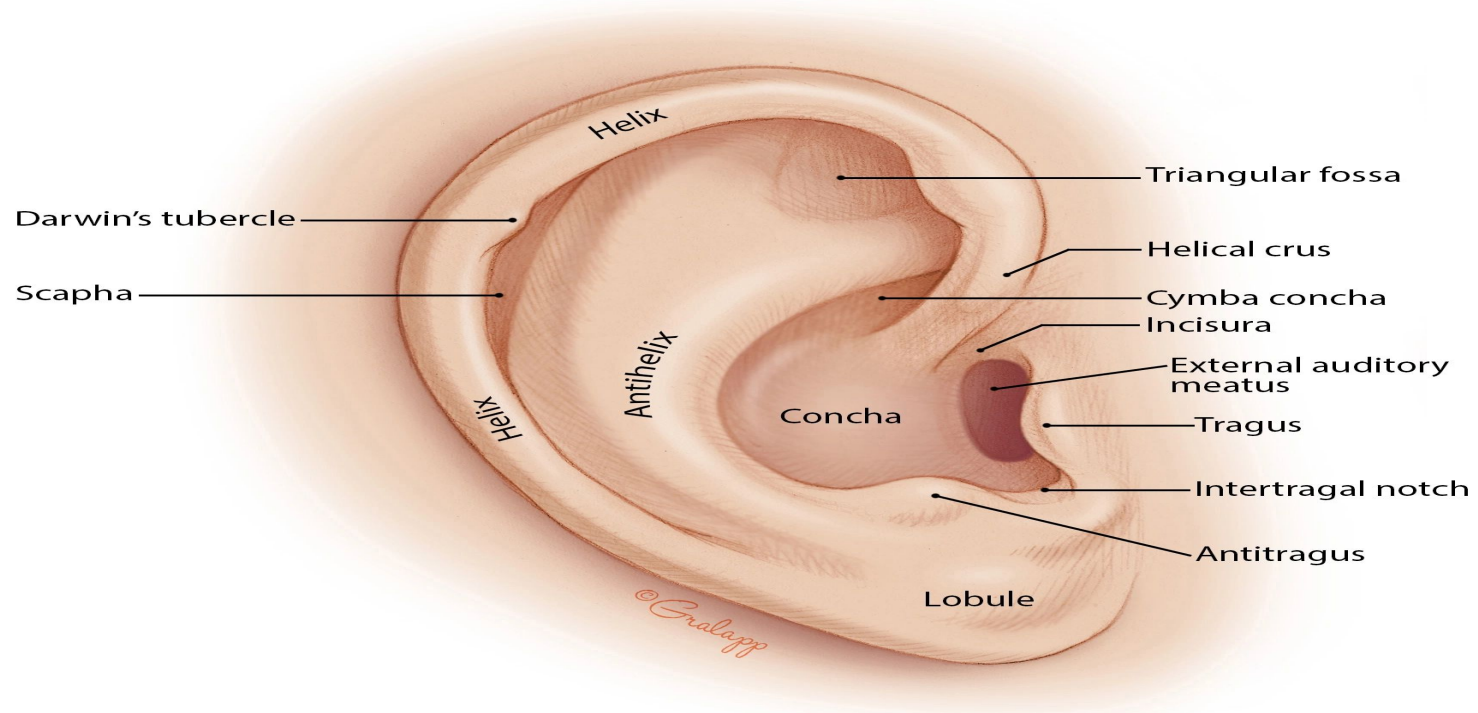
Cochlear
branch
(20%)

Post. Vestibular
A
(to saccule & PSSC)

PART	FUNCTION	DAMAGE	SENSORY ORGAN	
COCHLEA	HEARING	DEAFNESS	ORGAN OF CORTI	
UTRICLE/	LINEAR	ATAXIA	MACULAE	
SACCCULE	BALANCE			
SUP. SCC	ANGULAR	NYSTAGMUS	CRISTAE	
POST. SCC	BALANCE	VERTIGO	CUPULAE	
HORI. SCC				

CONGENITAL ABNORMALITIES OF COCHLEA

EXTERNAL EAR ANATOMY



HILLOCKS OF HIS COMBINE TO FORM PINNA

INCOMPLETE FUSION → PREAURICULAR SINUS



BAT EAR → ABSENT ANTI HELIX

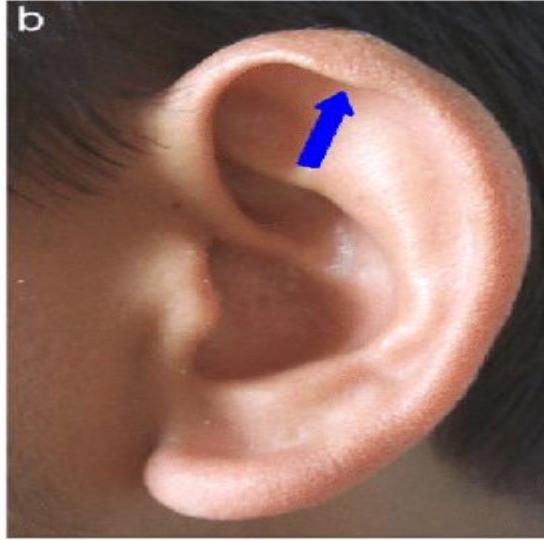
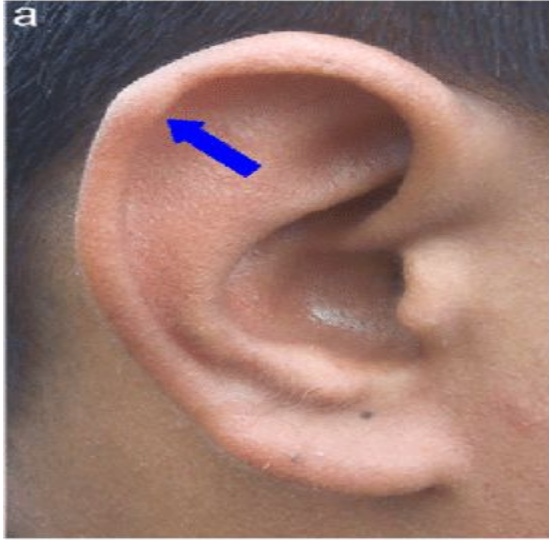
WILDERMUTH EAR → ABSENT HELIX

MOZART EAR -

MIXING OF HELIX AND ANTIHELIX

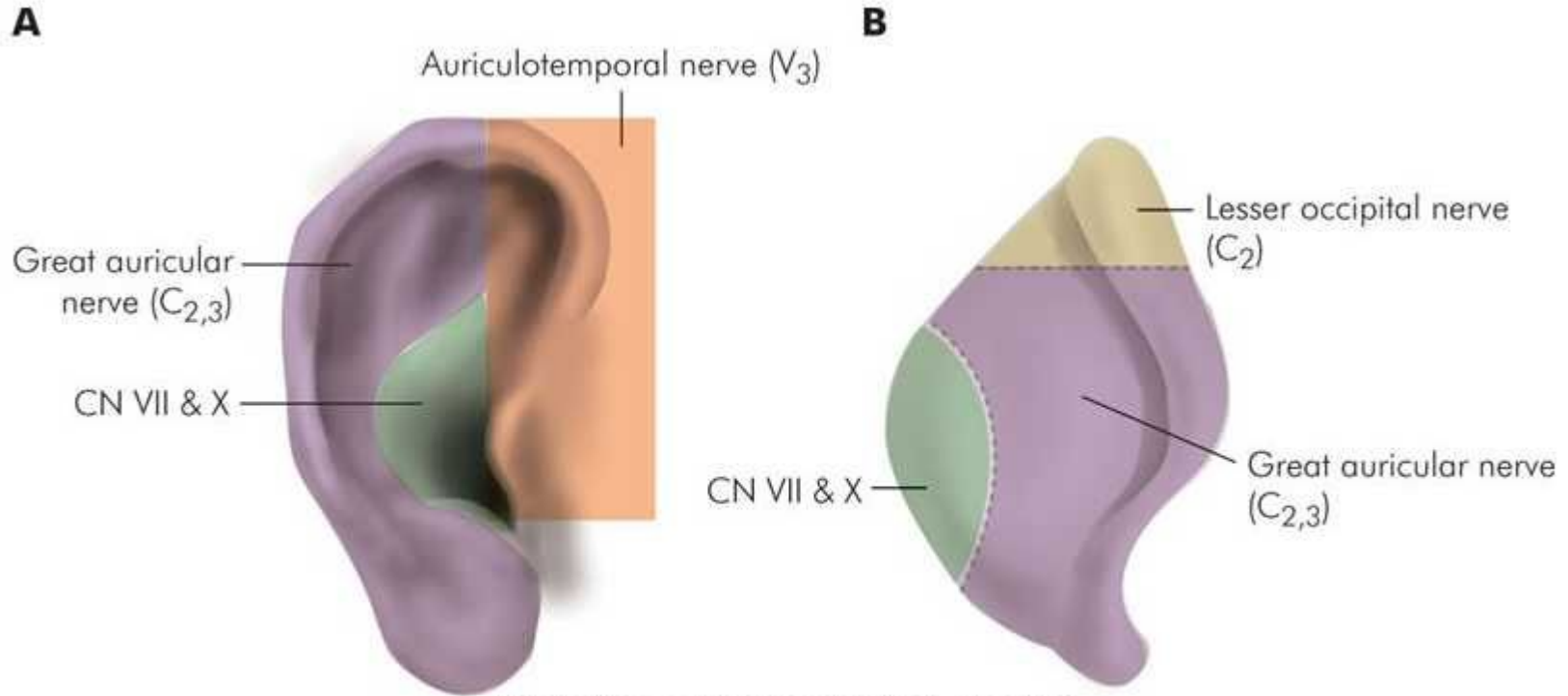


DARWIN TUBERCLE



**PINNA ACHIEVE ADULT SIZE BY 6 YEARS OF AGE SO NEVER
OPERATE BEFORE THAT**

NERVE SUPPLY OF PINNA



Dhingra: Diseases of Ear Nose and Throat, 5th Edition
Copyright © 2010 Elsevier. All rights reserved.

Figure 1.3 Nerve supply of pinna. (A) Lateral surface of pinna. (B) Medial or cranial surface of pinna.

EAC

LENGTH

ISTHMUS

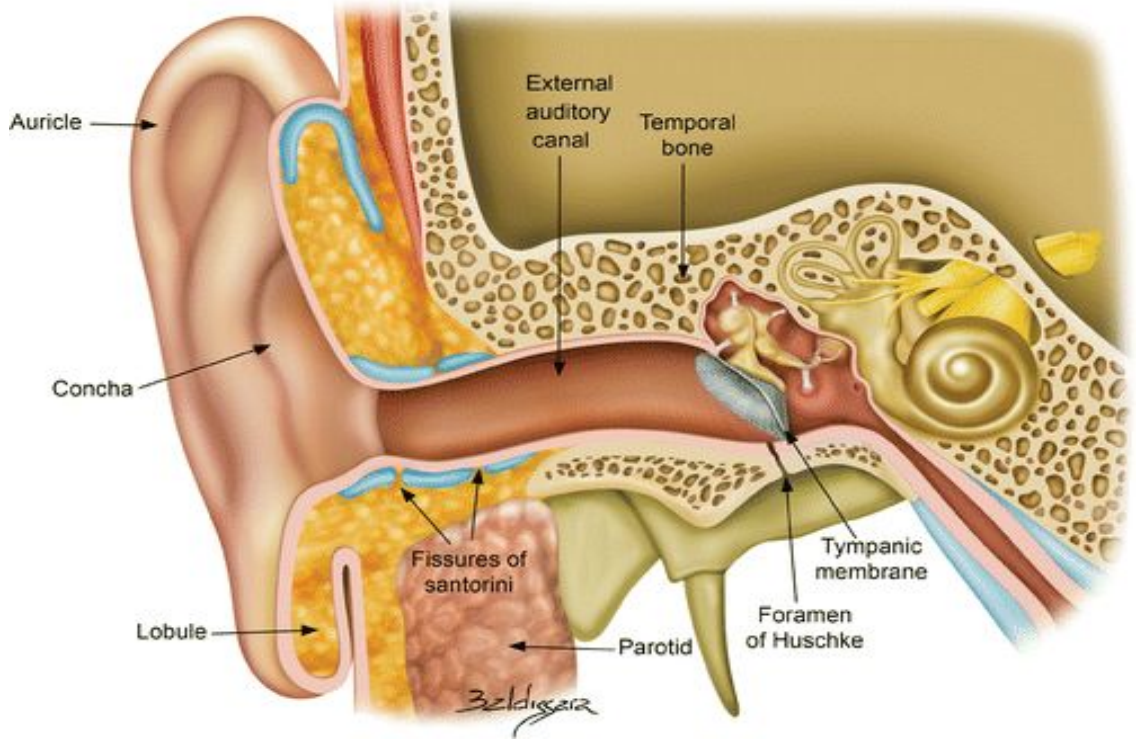
ORGANIC>>> INORGANIC

MOST DANGEROUS –

BUTTON BATTERY—>

FISSURES OF SANTORINI-
NORMAL (PAROTID)

FORAMEN OF HUSCHKE-
ABNORMAL (TM JOINT)



EUSTACHIAN TUBE

- Length of ET tube: 36mm
- Outer bony part is 12mm
- Inner cartilaginous part: 24mm



ET opens during swallowing. with the action of which muscle?

Ans: tensor palati muscle (tensor veli palati muscle)

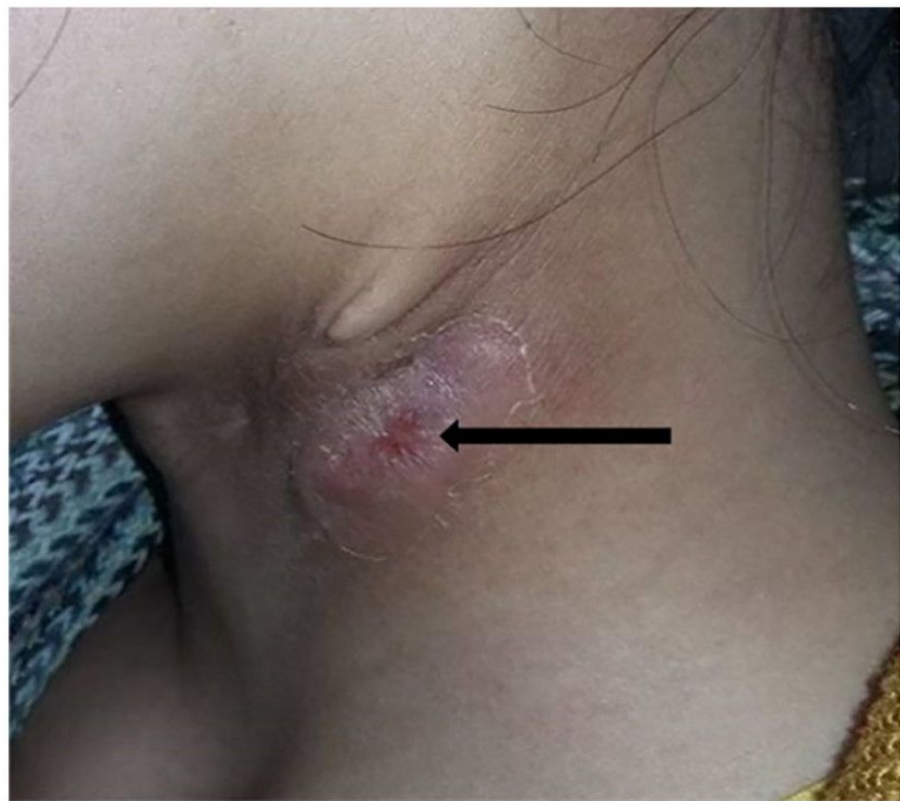
- Main function of ET: middle ear ventilation.
- If ET is blocked – negative pressure (vaccum) in middle ear – it will lead to RETRACTED TM

COLLAURAL FISTULA

ABNORMALITY OF THE FIRST BRANCHIAL CLEFT

ONE OPENING IN THE FLOOR OF EAC

SECOND OPENING BETWEEN SCM AND MANDIBLE IN NECK/ SOMETIMES
BEHIND THE MANDIBLE



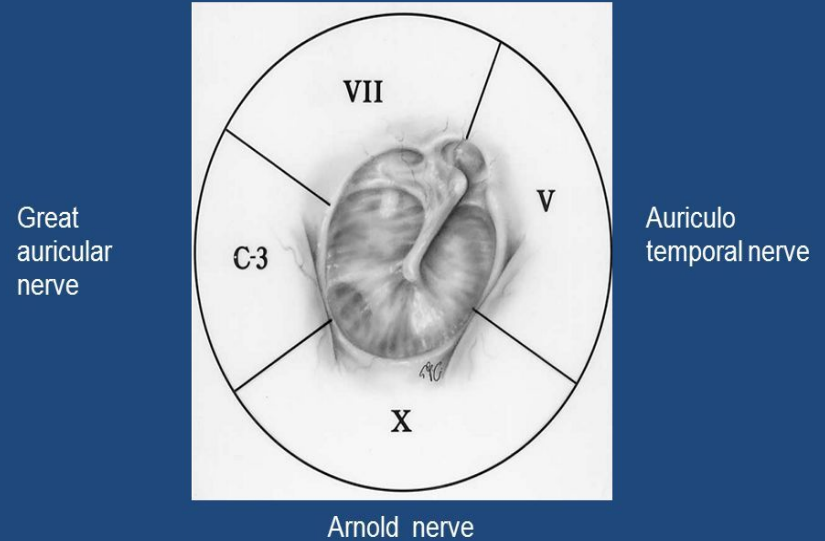
CLINICAL ASPECT

NR. OF WRISBERG(VII NR.) AND
HITZELBERGERGER SIGN

COUGH REFLEX IN AI PART DUE TO
VAGUS

LOSS OF SENSATION IN PI PART
DUE TO TB SPINE

Nerve supply of external auditory canal



TRANSFORMER MECHANISM/IMPEDANCE MATCHING MECHANISM OF MIDDLE EAR

AREA RATIO

OSSICULAR RATIO

CURVED MEMBRANE EFFECT

PHASE DIFFERENCE OF THE TWO WINDOWS

WHO CLASSIFICATION OF HEARING LOSS

AUDIOMETRY

FREQUENCY(Hz)

INTENSITY(dB)

AUDIOMETRIC ZERO-25dB

AUDIOMETRY

FREQUENCY(Hz)

INTENSITY(dB)

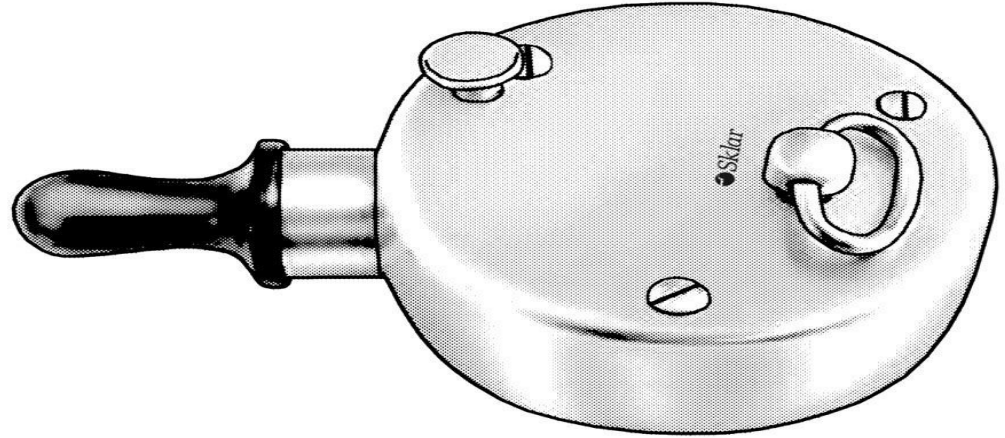
AUDIOMETRIC ZERO-25dB

DEAFNESS-
PT RESPONDS >25dB

IF NO RESPONSE AT ALL-
DEAD EAR

MASKING

MASKING DONE USING BARANY NOISE BOX



PINK NOISE /SOUND- MAINLY CONSISTENT SINGLE FREQUENCY

WHITE SOUND/NOISE- MULTIPLE FREQUENCY SOUND(SORT OF WIDE BAND)

LEVELS TILL WHICH YOU WILL SEE

CONDUCTIVE DEAFNESS

EXT. AND MIDDLE EAR

SENSORY DEAF.

COCHLEA

NEURAL DEAFNESS

RETROCOCHLEAR

(8TH NR AND BEYOND)

CHL	SNHL
Glue ear	Meniere's disease
CSOM	Acoustic neuroma (8 th nerve tumor)
otosclerosis	Drug induce hearing loss (ototoxicity)
	Age induced hearing loss (presbyacosis)

PTA COMPLETE

CARHARTZ NOTCH

ACOUSTIC DIP

RISING CURVE

DOWNSLOPING CURVE

U SHAPED/COOKIE BITE PATTERN- CONGENITAL DEAFNESS

OTOTOXIC DRUGS

AMINOGLYCOSIDES

CISPLATIN

DIURETICS

IN NEURAL DEAFNESS i.e RETROCOCHLEAR LESIONS

NERVE FATIGUABILITY TEST = TONE DECAY TEST= THRESHOLD DECAY
TEST= VERY POOR SDS SCORE

TUNING FORK



Tuning Fork

RINNE'S TEST

WEBER TEST

ABSOLUTE BONE CONDUCTION TEST

BING TEST - ALTERNATIVE TRAGUS PRESS → SNHL

GELLE TEST- SIEGELISATION → OTOSCLEROSIS

AVERAGE HEARING LOSS IN DIFFERENT LESIONS

COMPLETE OBSTRUCTION OF EAR CANAL → 30dB

PERFORATION OF TM (DEPENDS ON SIZE OF PERFORATION)-->10-40 dB

OSSICULAR INTERRUPTION WITH INTACT T.M →54dB

OSSICULAR INTERRUPTION WITH PERFORATION→ 38 dB

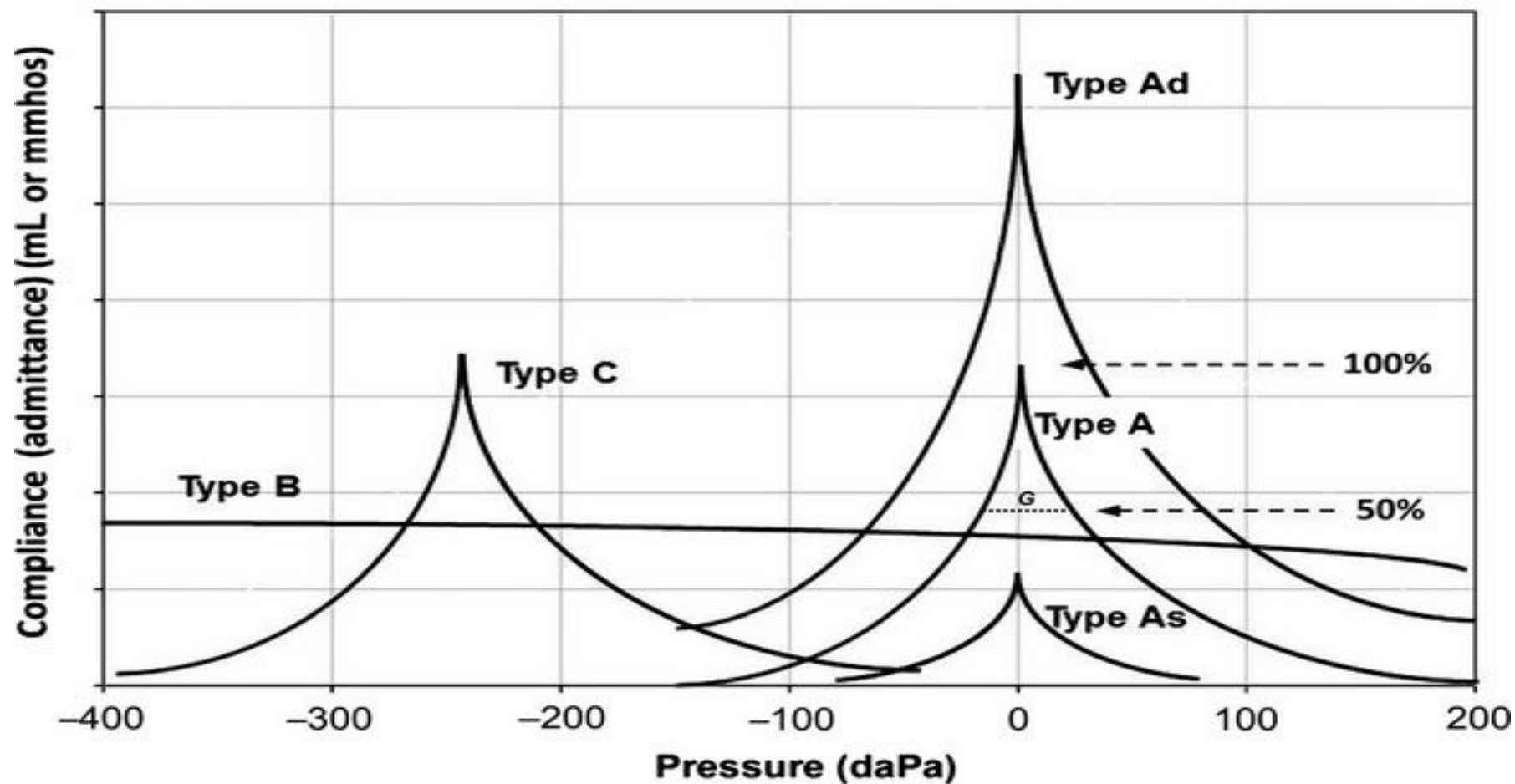
IMPEDENCE AUDIOMETRY

TYMPANOMETRY + STAPEDIAL REFLEX

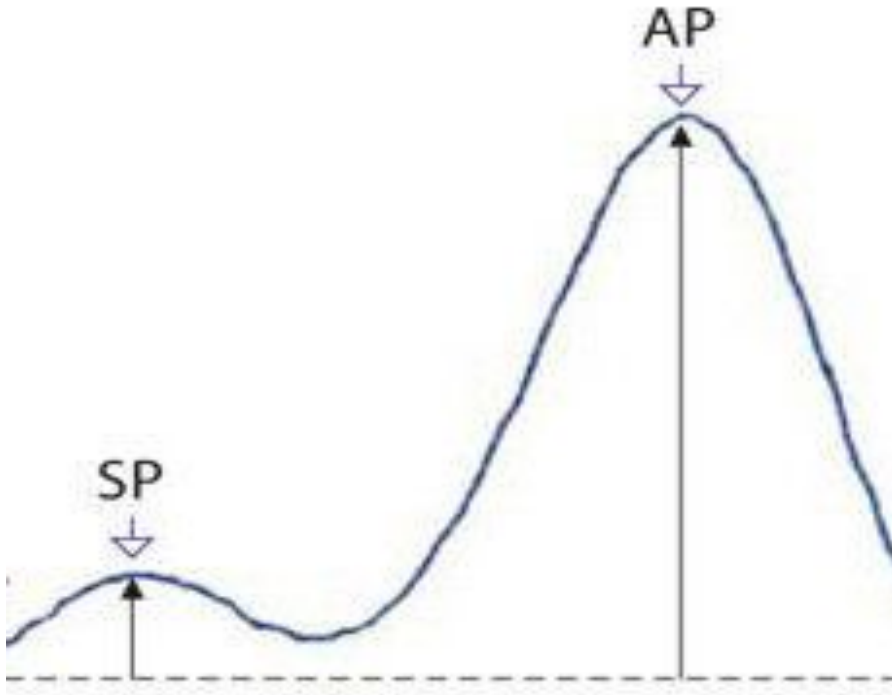
STAPEDIAL REFLEX

> 70 dB SOUND GIVEN → STAPEDIUS MUSCLE CONTRACT TO DAMPEN SOUND OR TO PROTECT

Tympanogram



ELECTROCOCHLEOGRAPHY

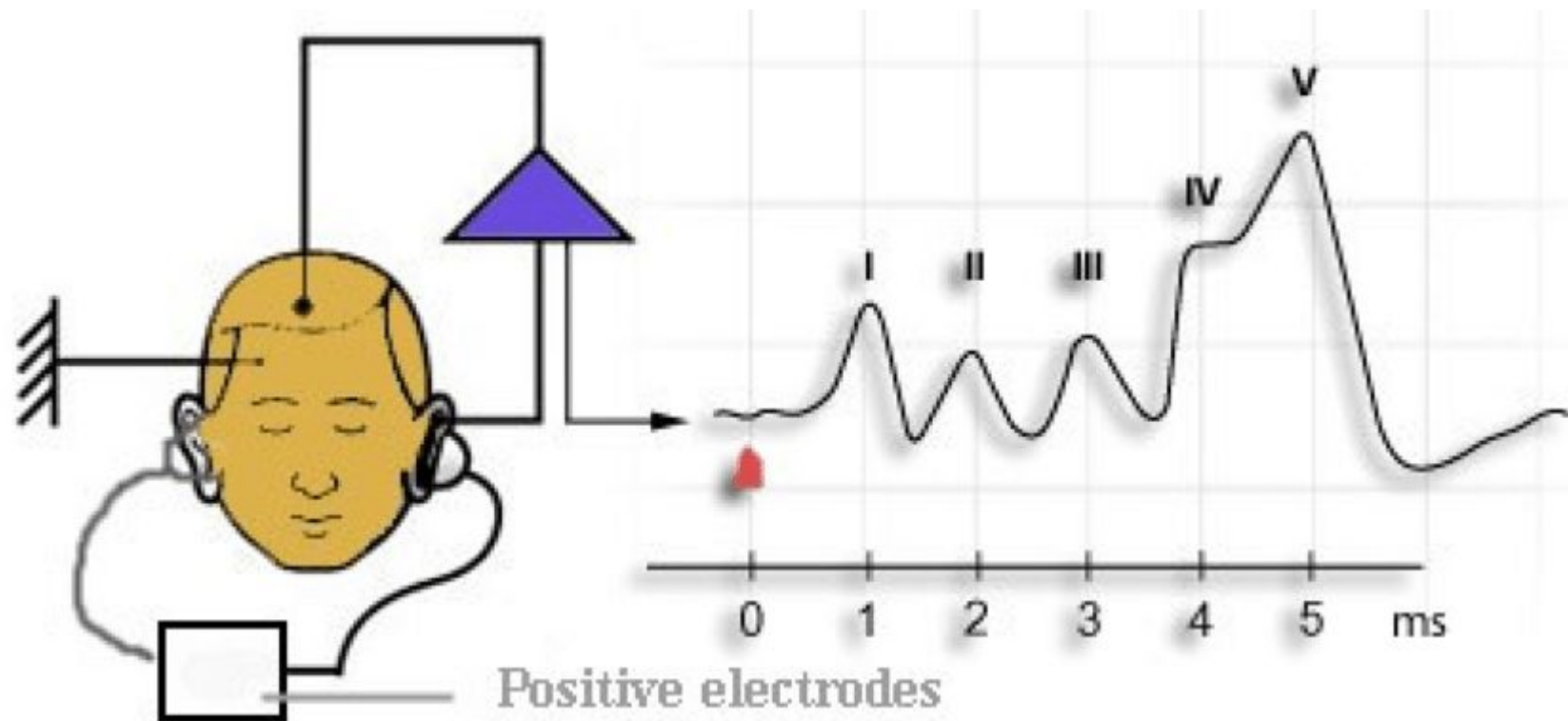


SP/AP RATIO

$< 0.3 \rightarrow$ NORMAL

$> 0.45 \rightarrow$ ABNORMAL

BERA



E- I, II

C-III

O-IV

L-V

I-VI,VII

V IS THE LARGEST WAVE

I IS MOST SENSITIVE WAVE

OTOSCLEROSIS

B/L

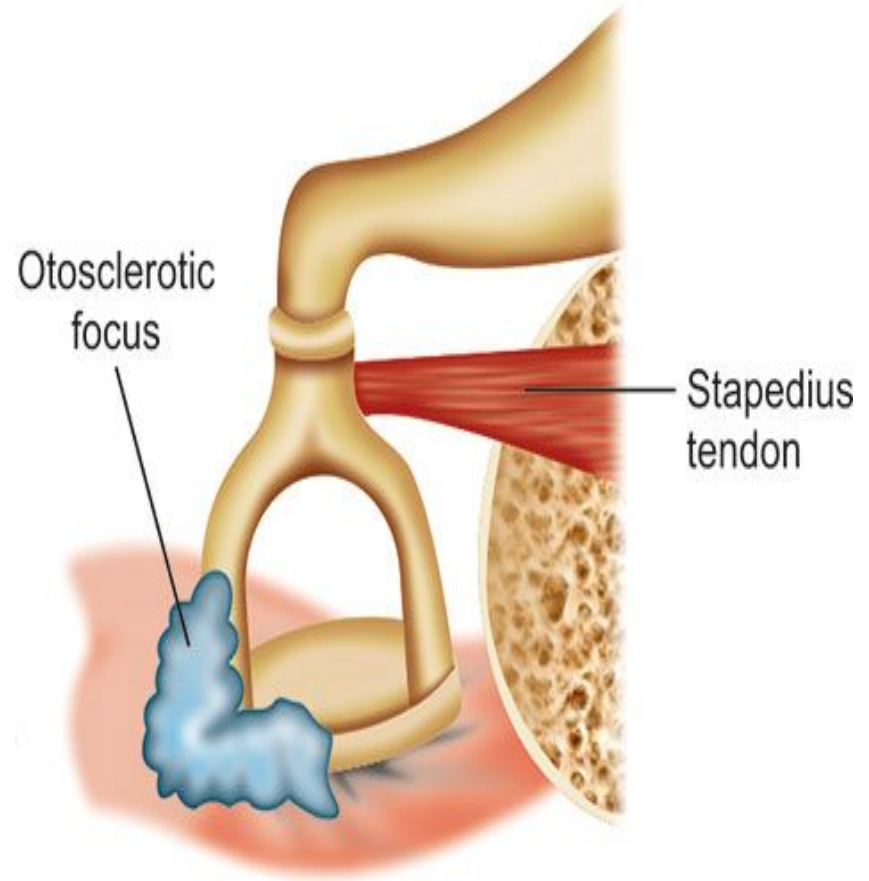
F>M

MOST IMP PREDISPOSING FACTOR

→ HEREDITARY

ASSOCIATED WITH PREGNANCY
AND MEASLES

VANDER HOEVE SYNDROME



CONDUCTIVE DEAFNESS- LOW INTENSITY /MILD TYPE

BILATERAL ,PAINLESS

IN PREGNANT FEMALES COMMONLY

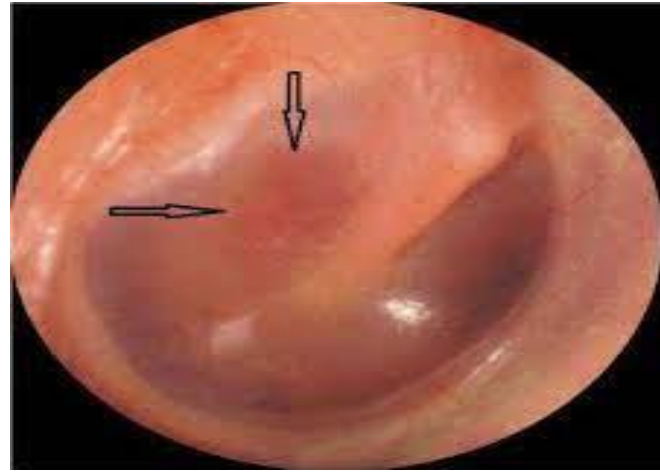
PARACUSIS WILLISI

TINNITUS AND VERTIGO LESS COMMON

SPEECH WILL BE SOFT

SCHWARTZ / FLAMINGO PINK SIGN

CARHARTZ NOTCH



IN ACTIVE / OTOSPONGIOTIC PHASE SODIUM FLUORIDE CAN BE GIVEN

STAPEDOTOMY -IT IS BETTER→ CARHARTZ NOTCH DISAPPEARS AFTER STAPEDOTOMY

STAPEDECTOMY

MENIERES DISEASE

UNILATERAL

ENDOLYMPHATIC HYDROPS

FLUCTUATING DEAFNESS

TVS

MODERATE HEARING LOSS

NO DECAY TEST- NO ROLL OVER PHENOMENON

RINNE- +VE

WEBER LATERALISED TO NORMAL EAR

IOC- ECOG

>0.45

LERMOYEZ/REVERSE MENIERES

TULIO PHENOMENON

TUMARKIN CRISIS

DIPLACUSIS WILLISI- PERCEPTION OF 2 SOUNDS AS ONE EAR IS NORMAL

RECRUITMENT PHENOMENON- 1dB CHANGE WILL BE FELT A BIG CHANGE

GLYCEROL TEST - GIVE ORALLY,ITS A DEHYDRATING AGENT THAT
REDUCES ENDOLYMPH PRESSURE→ IMPROVES HEARING

TREATMENT

MEDICAL

VESTIBULAR SUPPRESANTS MEDICATIONS

LOW SALT DIET

MENIETE DEVICE

DIURETICS

STEROIDS

ENDOLYMPHATIC SAC DECOMPRESSION(BELOW DONALDSON LINE)

LAST-GENTAMYCIN

ACOUSTIC NEUROMA/ VESTIBULAR SCHWANNOMA

M/C TUMOUR OF CEREBELLOPONTINE ANGLE

NERVE INVOLVED - INF. VEST. NR.>SUP. VEST. NR> COCHLEAR NR

M/C SYMPTOM - HEARING LOSS/DECREASED HEARING

SEVERE DEAFNESS(>80dB)

ROLL OVER + DECAY TEST +

IOC- BERA

IT CAUSES PRESSURE SYMPTOMS ON ALL CRANIAL NERVES EXCEPT I AND II CRANIAL NR

RADIOLOGICAL IOC- GADOLINIUM ENHANCED MRI

T/T - SURGERY

IF PATIENT DENIES SURGERY THEN GAMMA KNIFE RADIOTHERAPY

TRAUMATIC EAR PERFORATION MANAGEMENT



INFECTIONS OF EAR

OTITIS EXTERNA

OTITIS MEDIA

SEROUS OTITIS MEDIA (GLUE EAR) —> INFECTION OCCURS—> ASOM

M/C OR THE EARLIEST FEATURE OF ET DYSFUNCTION IS RETRACTED TM

TOSS CLASSIFICATION PARS FLACCIDA

- **Grade I:** The pars flaccida is not in contact with the **neck of the malleus**.
- **Grade II:** The retracted pars flaccida is in contact with the neck of the malleus and **clothing** it.
- **Grade III:** Here in addition to grade II features there is **minimal erosion** of the outer attic wall
- **Grade IV:** In this grade in addition to all the above said changes there is **severe erosion** of the outer attic wall or scutum.

SADE CLASSIFICATION

TABLE I.
Classification of Retraction Pockets of Pars Tensa as Proposed
by Sadé.

Stage	Description
1	Slight retraction of tympanic membrane
2	Retraction of tympanic membrane, touching the incus or stapes
3	Tympanic membrane touching the promontory
4	Tympanic membrane adherent to the promontory

which principle: Beer Can
SEROUS OTITIS MEDIA (GLUE EAR) AND ASOM ^{not}

STAGES OF ASOM

GROMMET

~~ASOM~~
ASOM
Grommet
+
Grommet
insertion.



COMPLETE CSOM - SAFE AND UNSAFE

RESERVOIR SIGN

WULLSTEIN CLASSIFICATION OF TYMPANOPLASTY

TYPE

I - MYRINGOPLASTY

II - MYRINGOINCUDOPEXY

III - MYRINGOSTAPEDOPEXY

IV - ON FOOT PLATE(MINOR/SMALL CAVUM EFFECT)

V - FENESTRATION SURGERY

OSSICULAR CHAIN RECONSTRUCTION

PORP

TORP

MATERIALS USED FOR OSSICULAR RECONSTRUCTION

TEMPORALIS FASCIA

PERICHONDRIUM

PERIOSTEUM

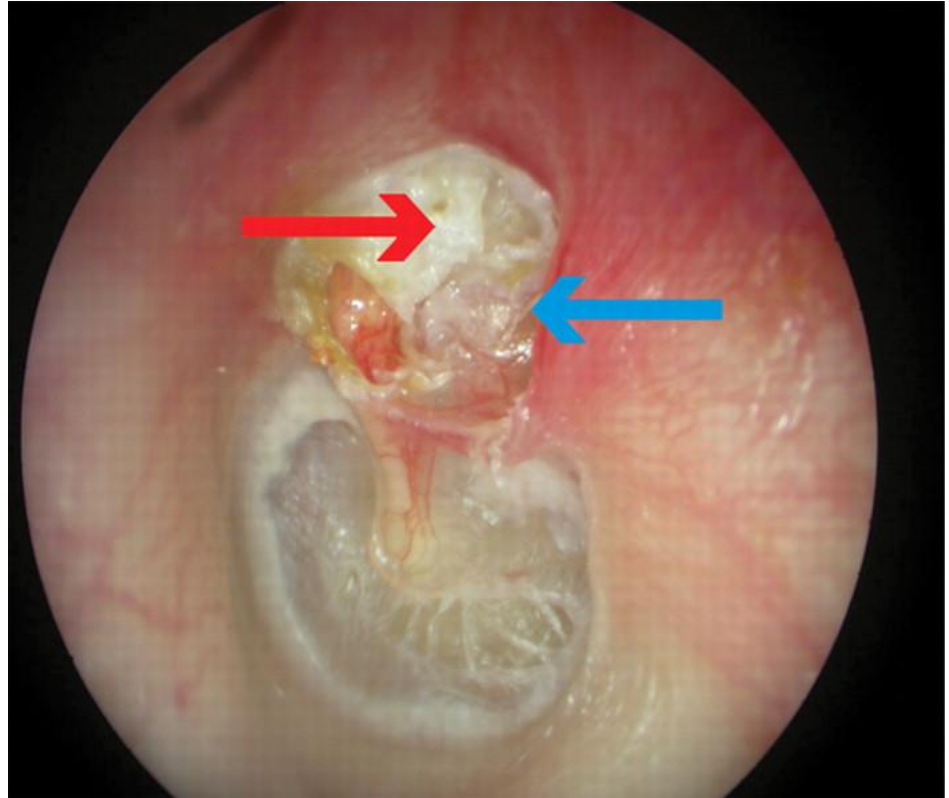
ARTIFICIAL GRAFT(ALLODERM)

CHOLEASTETOMA

METAPLASIA OF
THE ERODED EPITHELIUM

PRUSSACK SPACE-
M/C AND EARLIEST SITE

SINUS TYMPANI-
RESIDUAL AND REOCCURENCE



DX OF CHOLESTEATOMA

IT GENERALLY PRESENTS AS WHITE MASS BEHIND T.M

IT CAN BE:-

1. PAEDIATRIC- CONGENITAL CHOLESTEATOMA
1. RETRACTED T.M W/O INFECTION- PRIMARY CHOLESTEATOMA
1. H/O INFECTION, T.M PERFORATION - SEC. CHOLESTEATOMA
1. H/O ANY TRAUMA OR SURGICAL TRAUMA- TERTIARY CHOLESTEATOMA

HYPOTHESIS

RETRACTION POCKET FORMATION- WHITMAN'S THEORY

BASAL CELL HYPERPLASIA- RUEDIS THEORY

SQUAMOUS METAPLASIA- SADE THEORY

EPITHELIAL MIGRATION- HABERMAN THEORY

MASTOIDECTOMY TYPES AND STEPS

MASTOIDECTOMY TYPES AND STEPS

COMPLICATIONS OF CSOM

1. MASTOIDITIS

M/C OVERALL OR M/C EXTRACRANIAL-
MASTOIDITIS

M/C SYMPTOM OF MASTOIDITIS-
EAR DISCHARGE

M/C SIGN OF MASTOIDITIS- TENDERNESS

EARLIEST SIGN - IRONING OF MASTOID

HALLMARK SIGN- SAGGING OF CANAL



2. **ABSCESS**

BEZOLD

CITELLI

LUC ABSCESS

M/C IS POST AURICULAR ABSCESS

3. M/C PART OF THE FACIAL NR. INVOLVED IS TYMPANIC

4. LABYRINTHITIS

5. PETROSITIS- GRADIENGO TRIAD: 5TH(PAIN) AND 6TH(DIPLOPIA) NR PALSY WITH EAR DISCHARGE

REMEMBER ALSO

MAC EWANS TRIANGLE BOUNDARIES

SINODURAL ANGLE ALSO CALLED AS CITELLI ANGLE

INTRACRANIAL COMPLICATIONS

M/C MENINGITIS

2ND M/C BRAIN ABSCESS

SUBDURAL ABSCESS

OTIC HYDROCEPHALUS

LAST IMP COMPLICATION INTRACRANIAL-LATERAL SINUS THROMBOPHLEBITIS

HECTIC/PICKET FENCE FEVER

GRESINGER SIGN

CROWE BECK SIGN

TOBEY AYER SIGN

DELTA SIGN ON CT



INFECTIONS OF EXTERNAL EAR

BOXER EAR/ CAULIFLOWER EAR



DIFFUSE OTITIS EXTERNA NAMES

GENERALISED OTITIS EXTERNA

SWIMMERS EAR

SINGAPORE EAR

TROPICAL EAR

TELEPHONIC EAR

MALIGNANT OTITIS EXTERNA/ OSEOMYELITIS OF SKULL BASE

DIABETES PATIENT

PRESENTS AS EAR DISCHARGE + GRANULATION IN CANAL + CAN HAVE FACIAL PALSY

TC 99 SCAN FOR DIAGNOSIS

GALLIUM SCAN FOR PROGNOSIS

REMEMBER THESE DISEASES IN ENT FOR DIABETIC PATIENT

EAR- MALIGNANT OTITIS EXTERNA

NOSE- MUCORMYCOSIS (INVASIVE DISEASE, LIPOSOMAL AMPHOTERICIN B FOR TREATMENT)

ORAL CAVITY- LUDWIG ABSCESS

OSTEOMA AND EXOSTEOSIS DIFFERENCE

EXOSTEOSIS AKA SURFER EAR (FAST MOVING WATER)

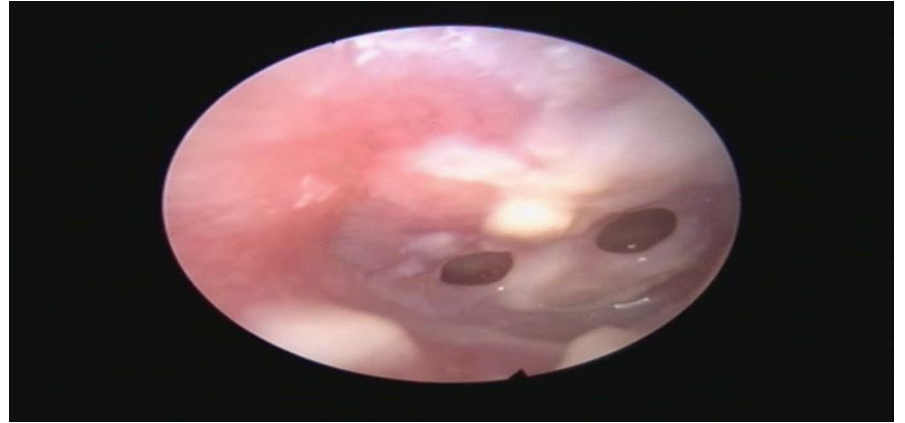
TUBERCULAR OTITIS MEDIA

PALE

PERFORATIONS(MULTIPLE)

PAINLESS

T/T - ATT



OTOMYCOSIS

WET NEWSPAPER LIKE APPEARANCE

ASPERGILLUS NIGER

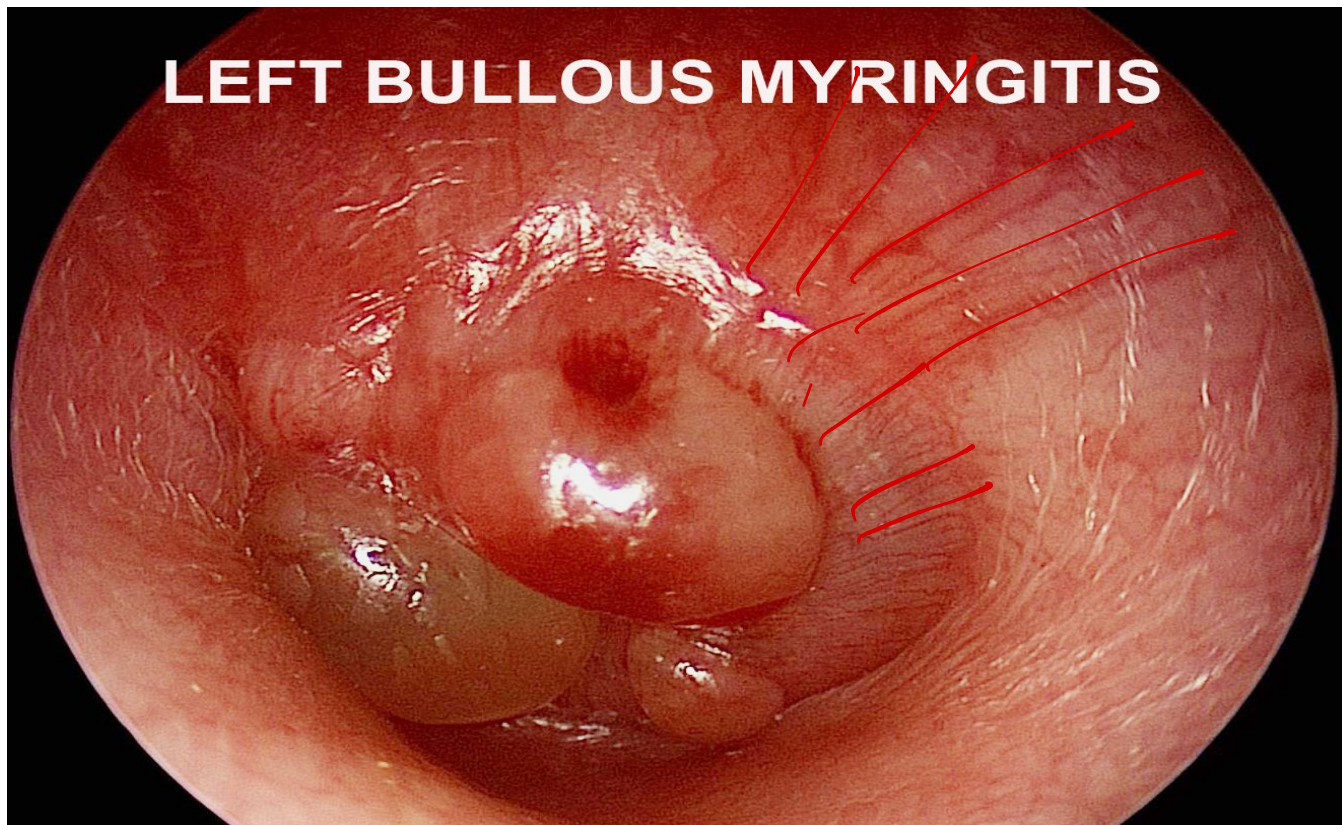


VIRAL INFECTIONS OF EXT. AND MIDDLE EAR

HERPES ZOSTER OTICUS—> IN IMMUNOCOMPROMISED PATIENTS



MYRINGITIS BULLOSA- PAINFUL BLEBS ON TM



NEWBORN SCREENING FOR HEARING

SCREENING PROCEDURES OF CHOICE

OAE- AFTER 24 HOURS/ 1ST DAY SHOULD BE DONE

SHOULD BE DEFINITELY DONE BY 1ST MONTH

BERA - BY 3RD MONTH

BUT REMEMBER

**FOR A NEWBORN IN ICU i.e HIGH RISK BABY - BERA IS THE SCREENING
PROCEDURE OF CHOICE DIRECTLY**

HEARING AID

COMPONENTS

R- RECEIVER

A- AMPLIFIER

M- MICROPHONE

USE 1. BTE- BEHIND THE EAR

2. CIC - COMPLETELY IN THE EAR

3. ITE- IN THE EAR



IF CANAL ATRESIA, CANAL STENOSIS, PUS IN THE CANAL PRESENT THEN BAHA IS USED

BAHA

1. Titanium implant



2. Abutment



3. Sound Processor



COCHLEAR IMPLANT

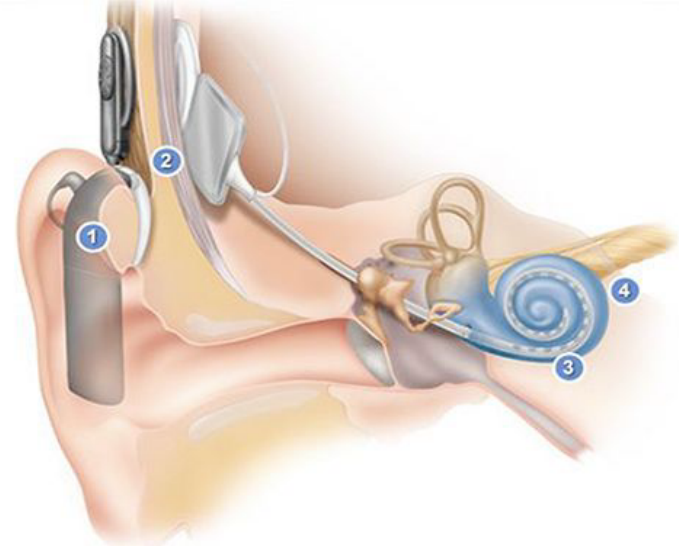
MONDINI DYSPLASIA IS
M/C INDICATION

MICHAEL APLASIA -

CONTRAINDICATED IN IT

POST LINGUAL DEAFNESS ARE BEST CANDIDATES

CAN BE USED IN PRE LINGUAL DEAFNESS ALSO



IN NEURAL DEAFNESS - BRAINSTEM IMPLANT IS USED - KEPT IN
LATERAL RECESS OF FOURTH VENTRICLE

FISTULA TEST

PRESS THE TRAGUS ALTERNATIVELY TO CHECK VESTIBULAR SYSTEM(NYSTAGMUS)

-VE IN NORMAL PEOPLE

+VE IN CHOLESTEATOMA

FALSE +VE MEANS SYMPTOMS +SENT IN ABSENCE OF FISTULA SEEN IN MENIERES DISEASE AND CONGENITAL SYPHILLIS- HENNEBERT SIGN

FALSE -VE MEANS SYMPTOMS -SENT IN PRESENCE OF FISTULA SEEN IN DEAD LABYRINTH AND EXTENSIVE CHOLESTEATOMA

→ Bx is C/I

~~[middle ear tumour]~~

Reddish mass behind intact
T.M.

GLOMUS TUMOUR/PARAGANGLIOMA

GLOMUS JUGULARE AND GLOMUS TYMPANICUM

SIGNS:

RISE SUN SIGN- TUMOUR COMING FROM BELOW

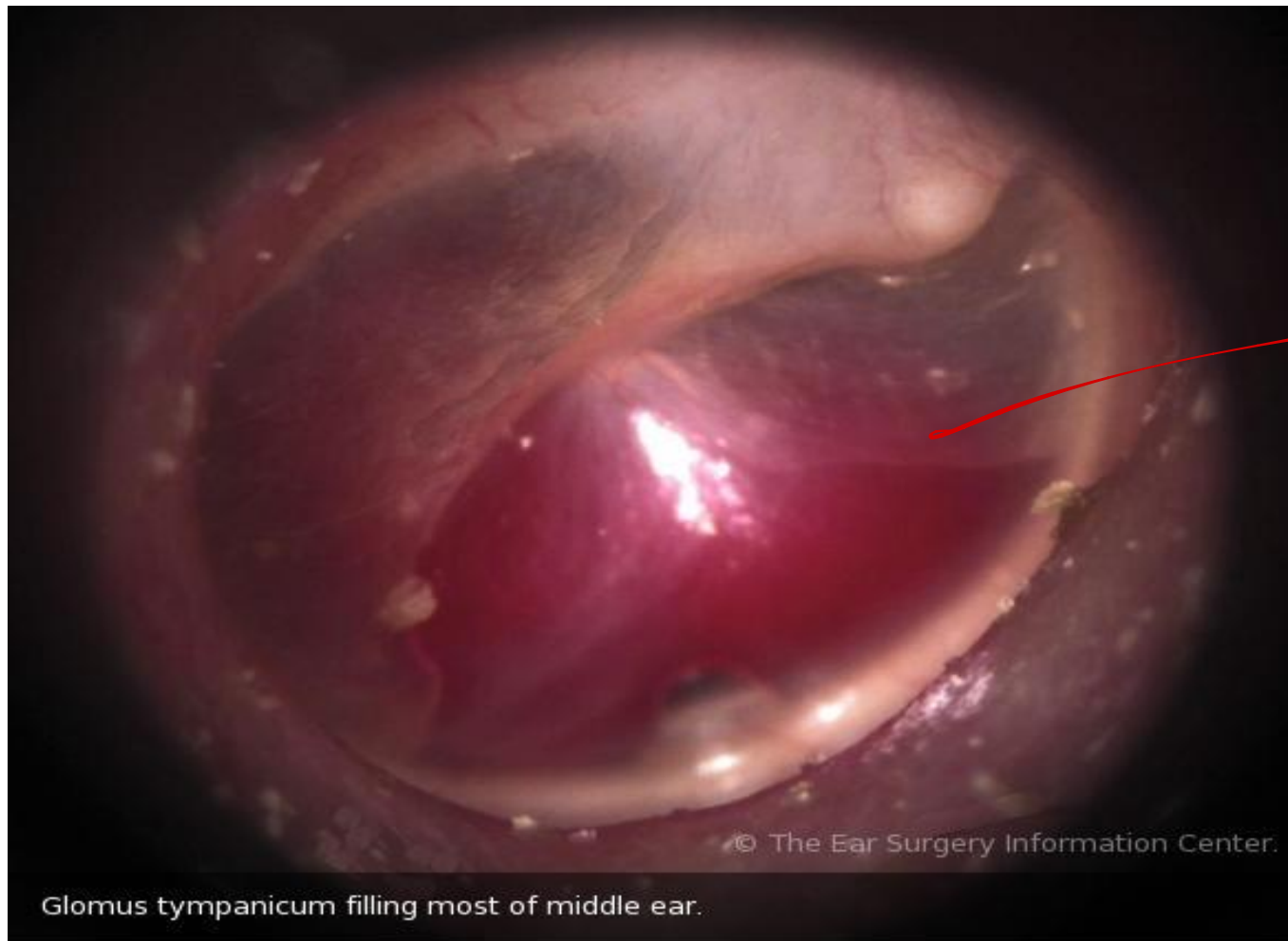
BROWN SIGN- BLANCHING ON PRESSING T.M (BY SIEGELISATION)

AQUINO SIGN- BLANCHING ON PRESSING NECK VESSEL BY
(SIEGELISATION)

CAROTID AND JUGULAR BONY PARTITION LOST- PHELPS SIGN

~~****~~
Pulsatile

Tinnitus



Pulsatile
tinnitus

© The Ear Surgery Information Center.

Glomus tympanicum filling most of middle ear.

SUDDEN SNHL

30dB

3 CONSECUTIVE FREQUENCIES

3 DAYS

CAUSES

M/C IS VASCULAR & IDIOPATHIC

T/T IS GENERALLY STEROIDS

CALORIC AND HALLPIKE TEST

CALORIC TEST

TEMP OF WATER IS 37 DEGREE CELSIUS

POSITION IS 60 DEGREE ANGLE AT BACK

IF NYSTAGMUS PRESENT IS NORMAL

IN MODIFIED KOBRAK CHILLED WATER IS USED

HALLPIKE (BITHERMIC CALORIC TEST)

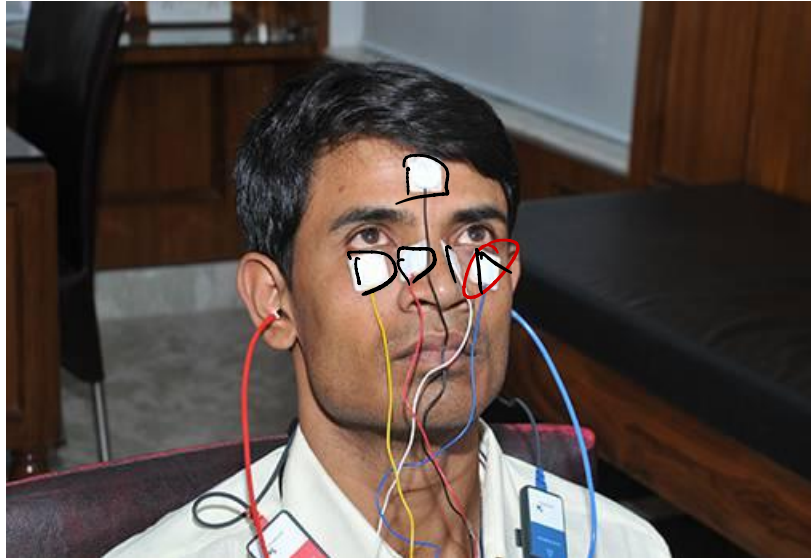
COWS

POSITION IS 30 DEGREE

VEMP

cVEMP- SCM MUSCLE- INHIBITORY RESPONSE-> INF VESTIBULAR NR.

oVEMP- EXTRAOCULAR MUSCLE- EXCITATORY RESPONSE-> SUP. VESTIBULAR NR.



BPPV

DUE TO OTOLITH / OTOCONIA MADE OF CALCIUM CARBONATE

MOSTLY POST. SCC

DIAGNOSIS BY -

DIX HALLPIKE MANUEVERE

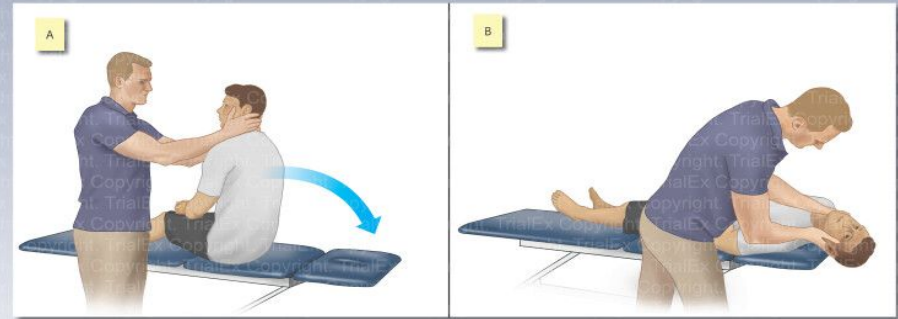
TREATMENT BY

EPLEYS MANUEVERE

SEMONT'S MANUEVERE

BRANDT AND DAROFF MANUEVERE

Dix-Hallpike Test



LARYNX

CONSISTS OF THREE PAIRED AND THREE UNPAIRED CARTILAGES

UNPAIRED

THYROID

CRICOID

EPIGLOTTIS

PAIRED

CUNEIFORM

CORNICULATE

ARYTENOIDS

THYROID ANGLE - ANGLE BETWEEN TWO THYROID LAMINA

PHARYNX

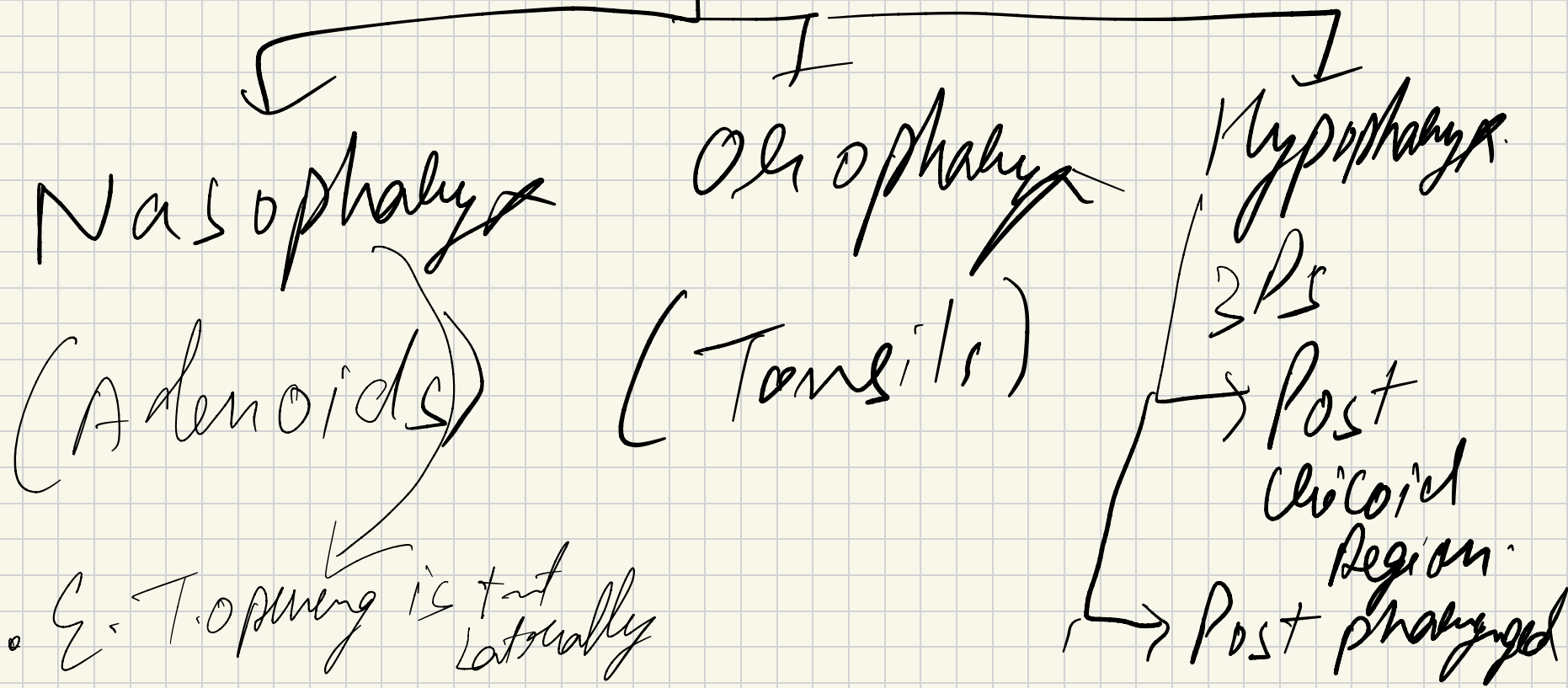
- It is a fibromuscular tube
- From skull base to c6 vertebrae
- It has 3 muscles
 1. Superior constrictor muscle (SC)
 2. Middle constrictor muscle (MC)
 3. Inferior constrictor muscle (IC)
 - a. Oblique fibers – thyropharyngeus (TP)
Circular fibers – cricopharyngeus (CP)
- Between skull base & superior constrictor there is space called: SINUS OF MORGAGNI (SOM)
- Eustachian tube passes through SOM.

Pharynx has 3 part:

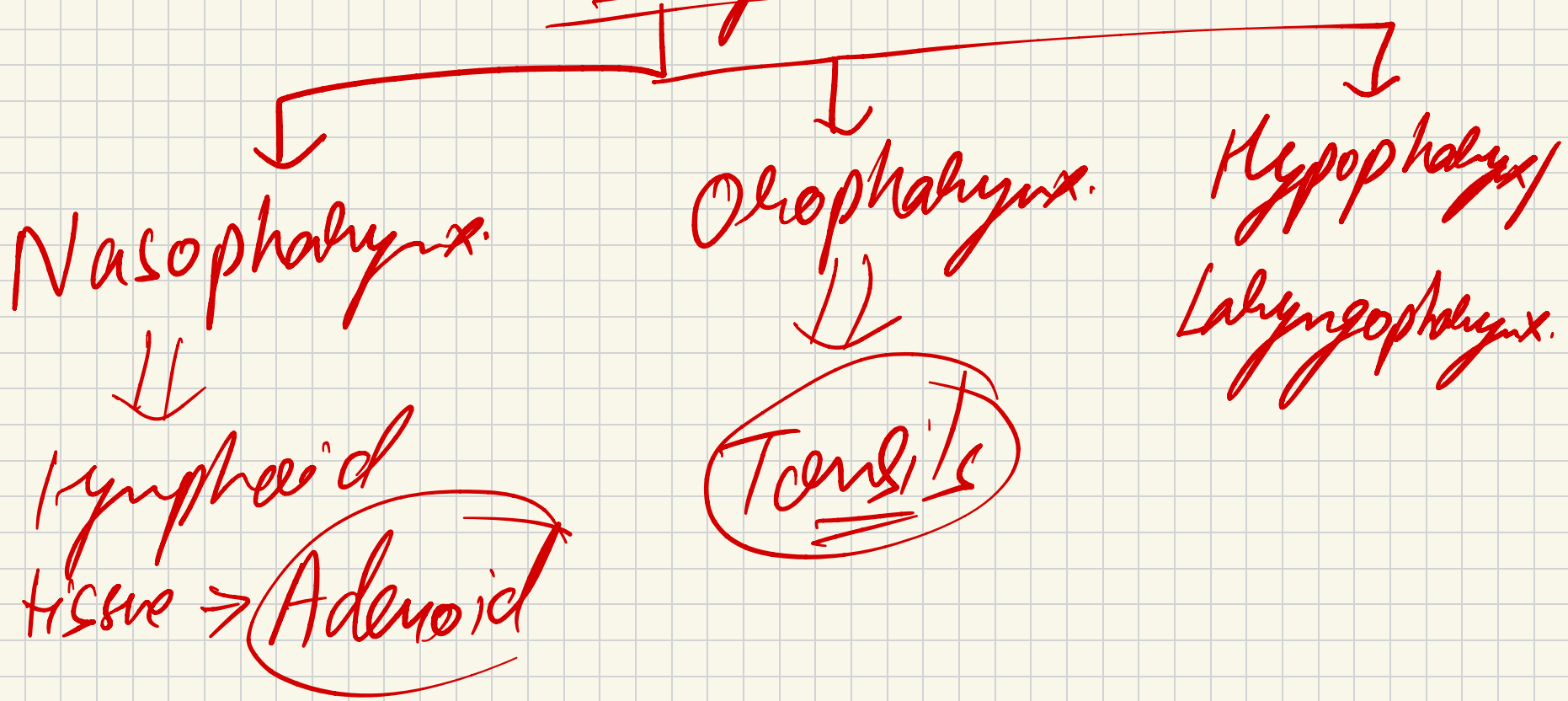
- I. Nasopharynx
 - Behind the nose
- II. Oropharynx
 - Behind the oral cavity
- I. Laryngopharynx
 - Behind the larynx
- LARYNGOPHARYNX Has 3 part:
 - a. Pyriform sinus (PS) (right & left)
 - b. Post cricoid area (PCA)
 - c. Posterior pharyngeal wall (PPW)

**Most common site of
hypopharyngeal
malignancy ? Ans:
pyriform sinus**

Pharynx

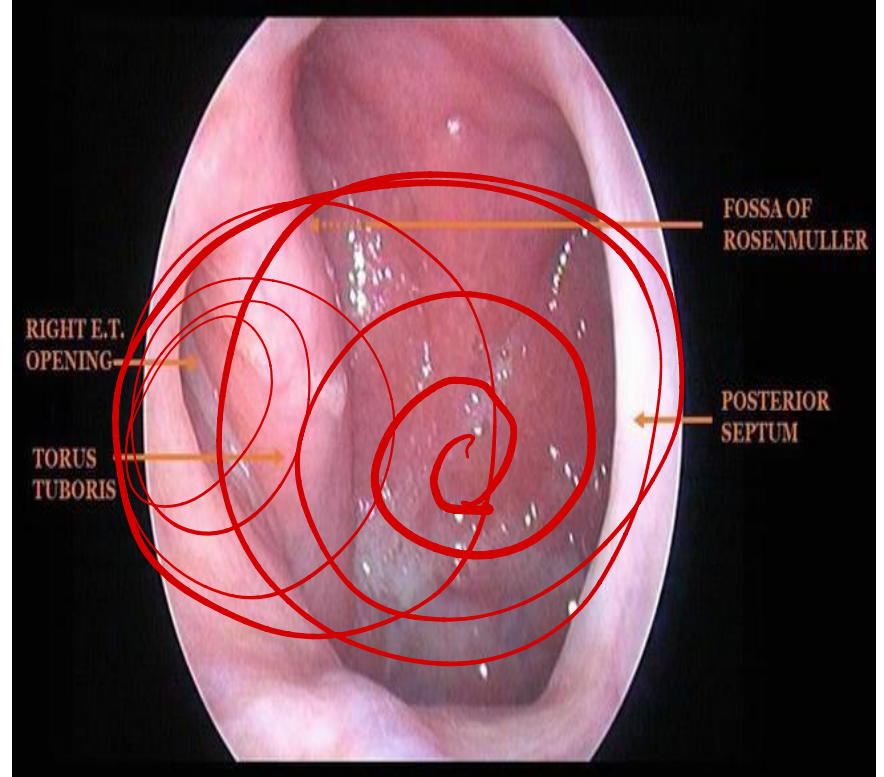


Pharynx.



ADENOIDS

→ lymphoid tissue of NPH



off : mouth breathing (snoring)
↓
heating.

OSA

Obstructive
sleep

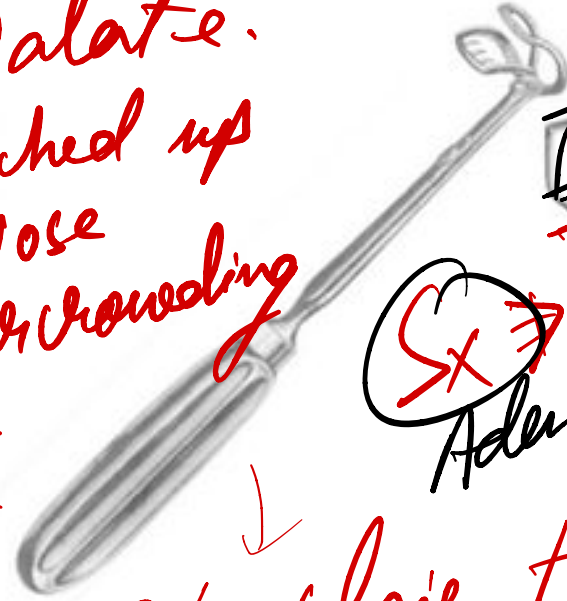
apnoea



Signs → 1. High arched
palate.

2. Pinched up
Nose

3. Overcrowding
of
teeth.



It

Sx → Adenoidectomy
Adenoidectomy.

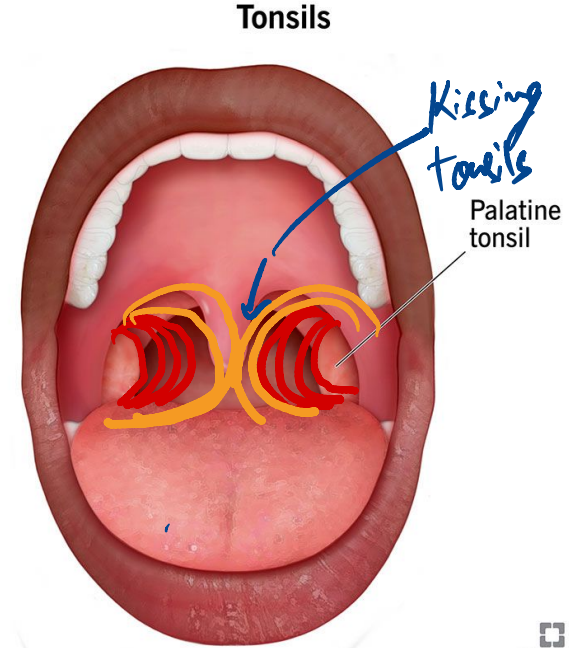
↓
St. Clair Thompson

If too much neck extension will lead to ATLANTOAXIAL SUBLUXATION (C1-C2)
this is called "Grisel syndrome")

TONSILS

BED OF TONSIL

- Is made by superior constrictor muscle.
- In the bed of tonsil lies:
 1. Styloid process (long and sharp bone).
 2. Glossopharyngeal nerve 9th nerve
- Long styloid process pressing 9th nerve – it will lead to throat pain refer to ear – this is called eagle syndrome (stylgia).



Pain →

Glossoph →

length of styloid process < 2.5 cm

Elongated styloid → > 4 cm

Haemorrhage

1° Hge.

- Bleeding during Sx.
- Palatoneilbar vein ligation. (T/T).

Reactive Hge

- < 24 hrs of Sx.
- Slippage of ligature
- Reexploration.

2° Hge
after 5 days of Sx

- Infections.
- iv antibiotics

Stygalgia. / Eagle Syndrome

Pain.

(N) length.

S.P. < 2.5 cm.

Styloid
process

Elongated S.P. > 4 cm.

Glossopharyngeal Ns.



Ludwig Angina

Rapidly spreading cellulitis
of submandibular spaces.

→ Etiology

└ Poor oral/dental hygiene.
└ Diabetes.

$T/E \Rightarrow I \& D$

Complication of Tonsillectomy.

Haemorrhage

1° Hge.
⇒ Bleeding during
Sx.

⇒ Palatonsillar
ven.

Reactionary

⇒ within 24
hrs of Sx.

⇒ Slippage of
suture

2° Hge

⇒ after 5-7
days.

⇒ Infection.

~~TH~~
⇒ ligate/
canterize.

~~TH~~
⇒ Reexploration. → i.v antibiotics.

process.

Most important source of blood supply – **tonsillar branch of facial artery**

• Venous drainage of tonsil: **paratonsillar vein** – it is main source of bleeding during tonsillectomy

Tonsillectomy.

HEMORRHAGE IN TONSILLECTOMY

• Has 3 type:

1. Primary – during Sx
2. **Re**actionary – post op. It is within 24 hr of Sx
Cause: slippage of ligature. It is an emergency
Rx: immediate **re**-exploration. ^Q
3. Secondary – post op. After 5th day of Sx. Cause: infection
Mild bleeding only Rx: IV antibiotics ^Q

1st → Mx complication of tonsillectomy is Bleeding.

Haemorrhage

1°

Occurring during
Sx

→ Paratonsillar vein

T/t → ligation/
cauterization

Reactionary

↓
within 24 hrs
of Sx

→ Slippage of ligature

T/t + Reexploration

2°

After 5 days
of Sx

→ Infection

→ iv
antibiotics

CAUSES OF WHITISH MEMBRANE ON TONSIL

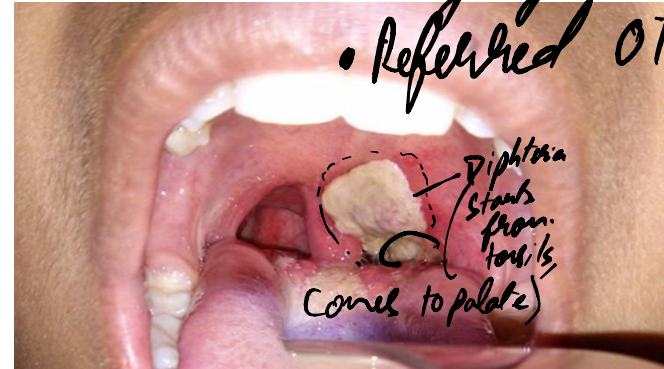
M/C/C- ACUTE MEMBRANOUS
TONSILLITIS

INFECTIOUS MONONUCLEOSIS
BY EBV

DIPHThERIA-EXTENDS BEYOND
TONSILS TO PALATE

CANDIDIASIS

VINCENTS ANGINA



- Styalgia
- Referred otalgia

Diphtheria
(Starts from
tonsils
comes to palate)

QUINSY (PERITONSILLAR ABSCESS)

- It is collection of pus between tonsil and its bed.
- It's more common in adult
- Unilateral
- Examination: uvula & tonsil push to other side.

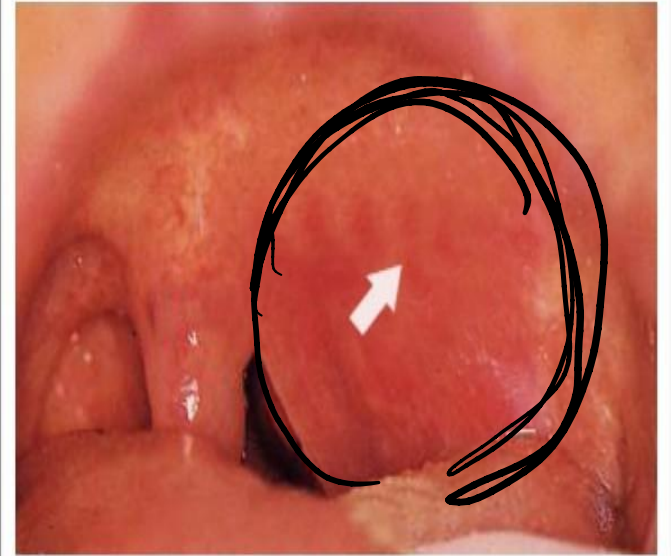
IB 2

QUINKES DISEASE

↳ edema of uvula



PERITONSILLAR ABSCESS (QUINSY)

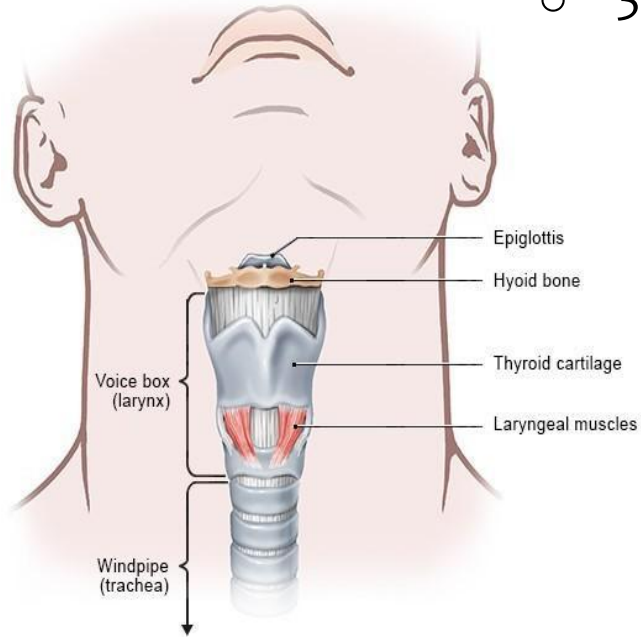


Ludwig Angina

- Diabetic patients.
- Rapidly spreading cellulitis of submandibular spaces
- T/t: I & V + Oral hygien.
+ Control/diabetes.

LARYNX

- Larynx is made from 6 cartilages
 - 3 are unpaired
 - 3 are paired



Laryngeal Cartilage's

- 3 Unpaired Cartilage's

-Epiglottis

-Thyroid

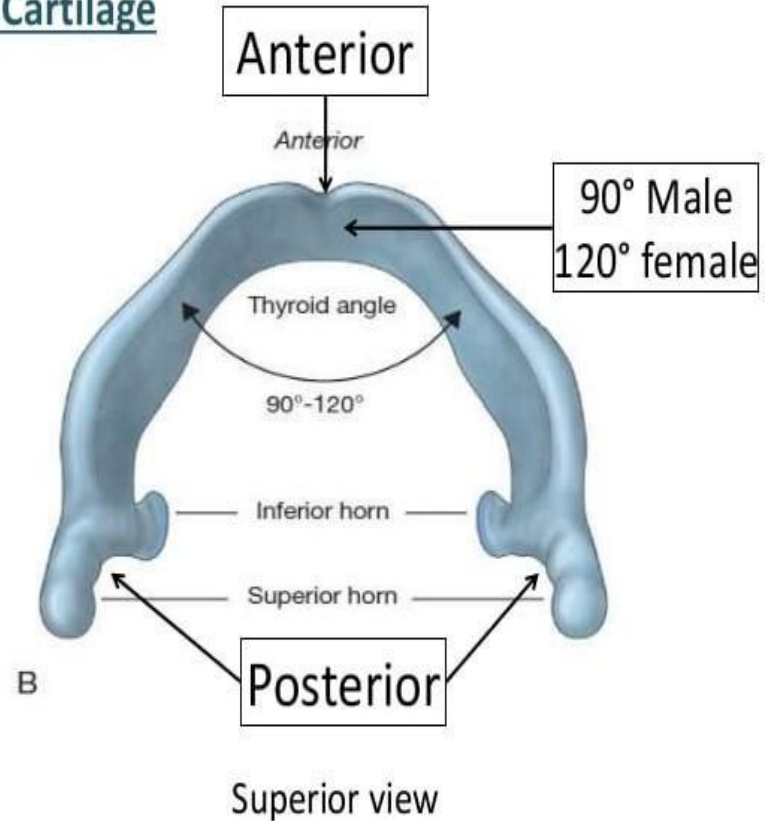
-Cricoid



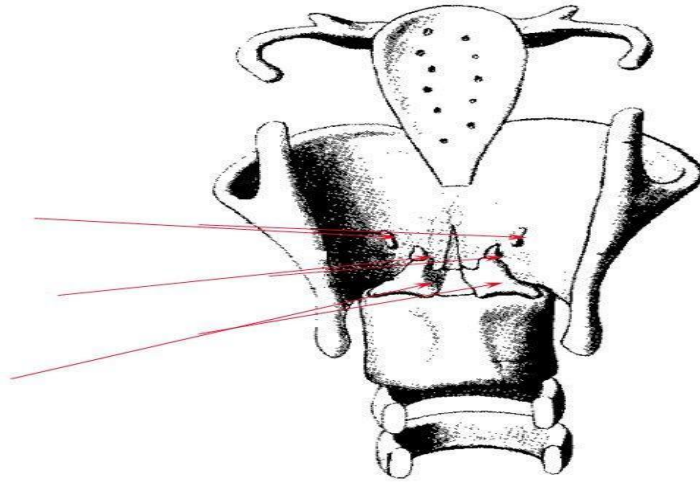
Thyroid cartilage

- It's open book like cartilage
- It's 2 laminae
- Angle between 2 laminae of thyroid cartilage called: thyroid angle
- In male: 90° (more sharp)
- In female: 120° (flat)

Thyroid Cartilage



- 3 Paired Cartilage's
 - Cuneiform
 - Corniculate
 - Arytenoid



11

LARYNX IS LINED BY CILIATED COLUMNAR EPITHELIUM EXCEPT VOCAL CORDS THAT ARE LINED BY STRATIFIED SQUAMOUS EPITHELIUM

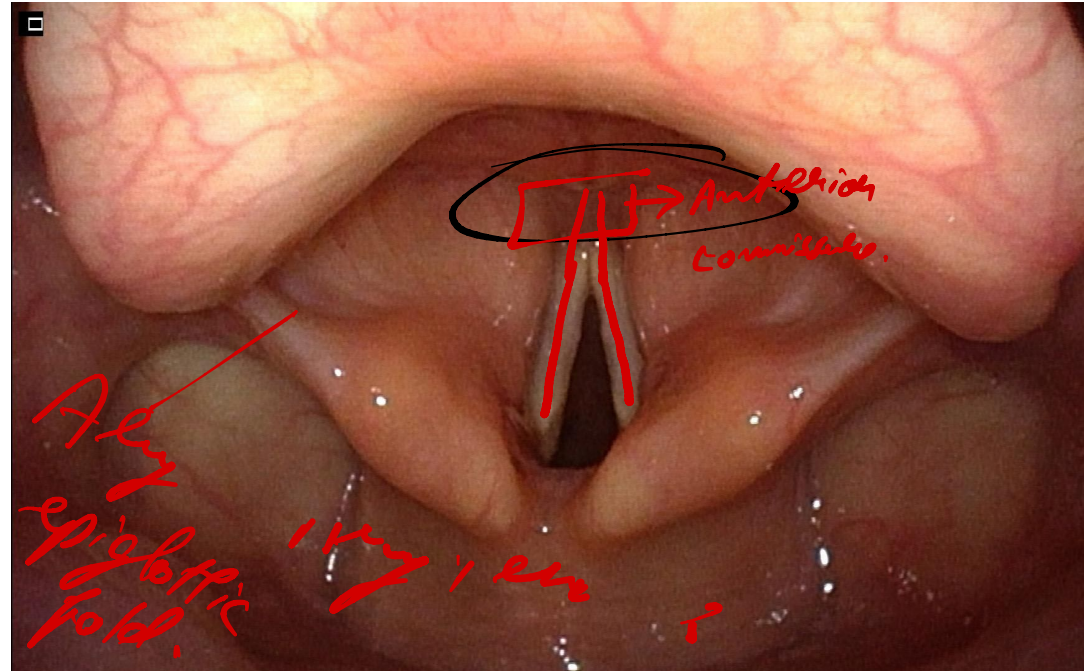
THREE DIVISIONS OF LARYNX

SUPRAGLOTTIS- EPIGLOTTIS, AE FOLDS, FALSE VC, VENTRICLES, SACCULE

GLOTTIS-

TRUE VOCAL CORD

SUBGLOTTIS



Laryngeal crepitus (Cricovertebral crepitus, Muir's crackle)

- ▶ The grating sensation produced by the movement of cricoid cartilage over the cervical vertebra (because cricoid is complete ring among the laryngeal cartilages). Hence Laryngeal crepitus is also called crico-vertebral crepitus.
- ▶ How to elicit laryngeal crepitus
 - ▶ It is produced when the larynx is moved sideways with slight posterior pressure.
- ▶ Normally laryngeal crepitus is present
- ▶ Absent laryngeal crepitus is Bocca's sign or Trotter's sign

LENGTH OF VOCAL CORD

MALE- 18-23 mm

FEMALE-16-17 mm

→ Laryngeal Calcification.

MALES HAVE LOW PITCH(DULL) VOICE

FEMALES HAVE HIGH PITCH(SHARP) VOICE

DISORDERS

PUBERPHONIA- FEMININE VOICE IN ADULT MALE

ANDROPHONIA- MASCULINE VOICE IN FEMALE

REINKES EDEMA

CAUSE

SMOKING

VOICE ABUSE

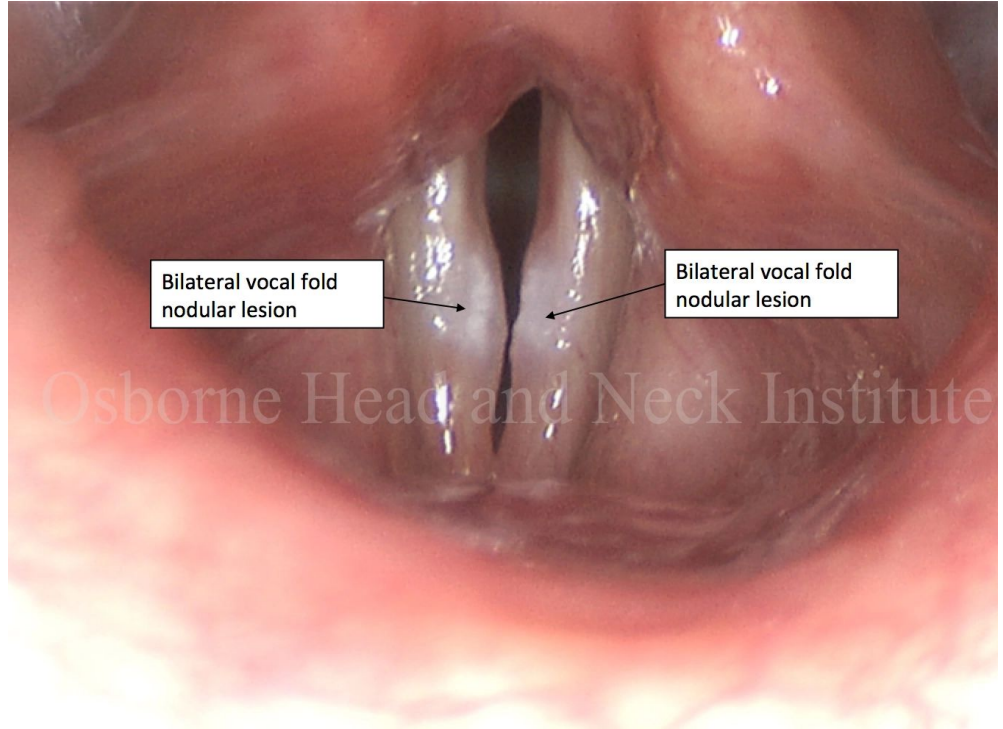


VOCAL NODULE/SCREAMER NODULE/TEACHER NODULE

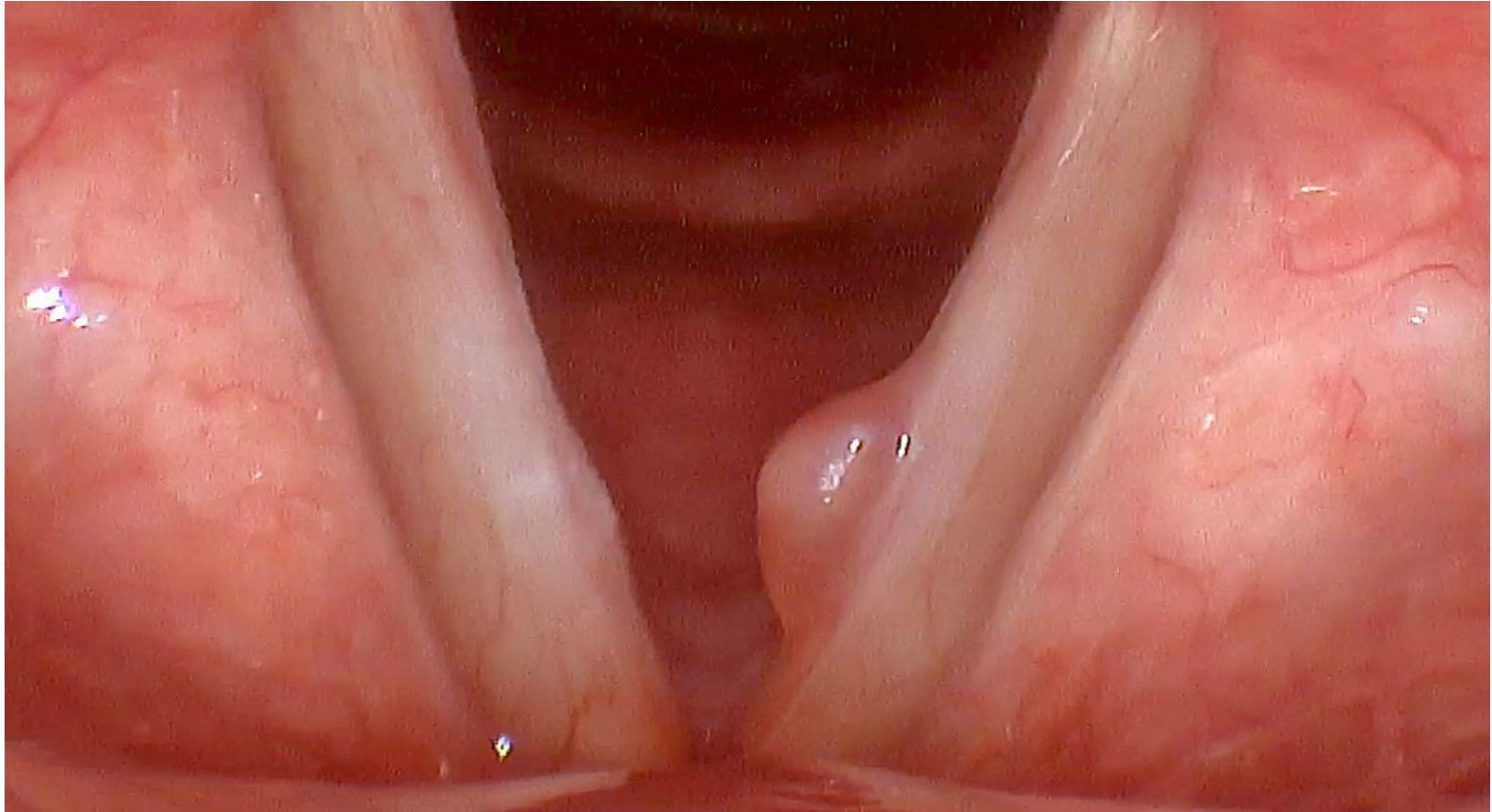
CAUSE- VOICE ABUSE

LPR

ALWAYS BILATERAL



VOCAL POLYP



Pediatric Vs. Adult

**Significant Anatomical Airway Differences between a Pediatric Patient
and an Adult Patient**

Airway	Pediatric	Adult
Tongue	Large relative to oropharynx	Normal
Epiglottis shape	Floppy, short and omega shaped	Firm, flatter
Epiglottis level	Larynx higher at C3-C4	C5-C6
Larynx shape	Funnel shaped	Columnar/cylindrical
Narrowest portion	At cricoid cartilage (subglottic)	At vocal cords

METHOD OF EXAMINATION OF LARYNX



IDL DOESN'T SHOW

- LARYNGEAL SURFACE
OF EPIGLOTTIS
- ANT.COMMISURE
- VENTRICLES
- UNDERSURFACE OF
VOCAL CORDS

FUNCTIONS OF LARYNX

PROTECTION OF AIRWAY

SOUND PRODUCTION

STRIDOR- NOISY BREATHING DUE TO AIRWAY OBSTRUCTION

INSPIRATORY-SUPRAGLOTTIS

EXPIRATORY- TRACHEA AND LUNGS

BIPHASIC-GLOTTIS, SUBGLOTTIS

MUSCLES OF LARYNX

ADDUCTORS

CRICOTHYROID(MAIN TENSOR ALONG WITH VOCALIS MUSCLE)

THYROARYTENOID

LATERAL CRICOARYTENOID

INTERARYTENOID

ABDUCTOR

PCA(SAFETY MUSCLE OF LARYNX)

MOTOR SUPPLY NERVE OF LARYNX

ALL MUSCLES ARE SUPPLIED BY RLN EXCEPT CRICOTHYROID WHICH IS SUPPLIED BY EXT BRANCH OF SLN

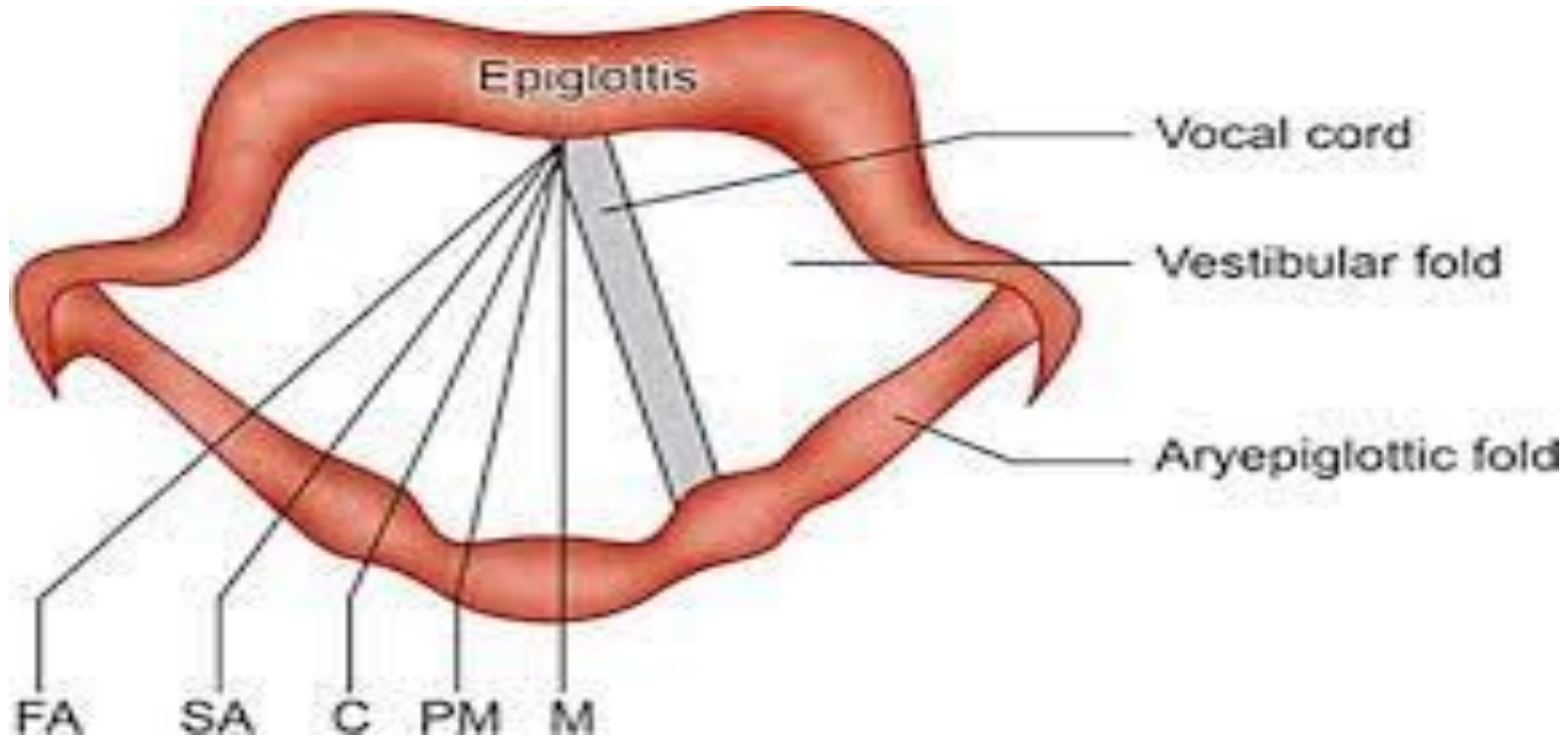
SENSORY SUPPLY OF LARYNX

SUPRAGLOTTIS - INTERNAL BR. OF SLN

GLOTTIS - MAINLY RLN/ ALSO INTERNAL BR. OF SLN

SUBGLOTTIS-RLN

0,1.5,3,7,9 mm POSITIONS OF VC



ONE LINERS

INCIDENCE OF INJURY RLN> >>SLN

LEFT RLN >> RIGHT RLN

IN THYROID SURGERY- EXT. BRANCH OF SLN

M/C/C OF U/L VOCAL CORD PALSY- MALIGNANCY(BRONCHUS CA)

M/C/C OF B/L VOCAL CORD PALSY- THYROID SURGERY

VOCAL CORD PALSY

U/L RLN(ABDUCTOR) PALSY

B/L RLN (ABDUCTOR) PALSY

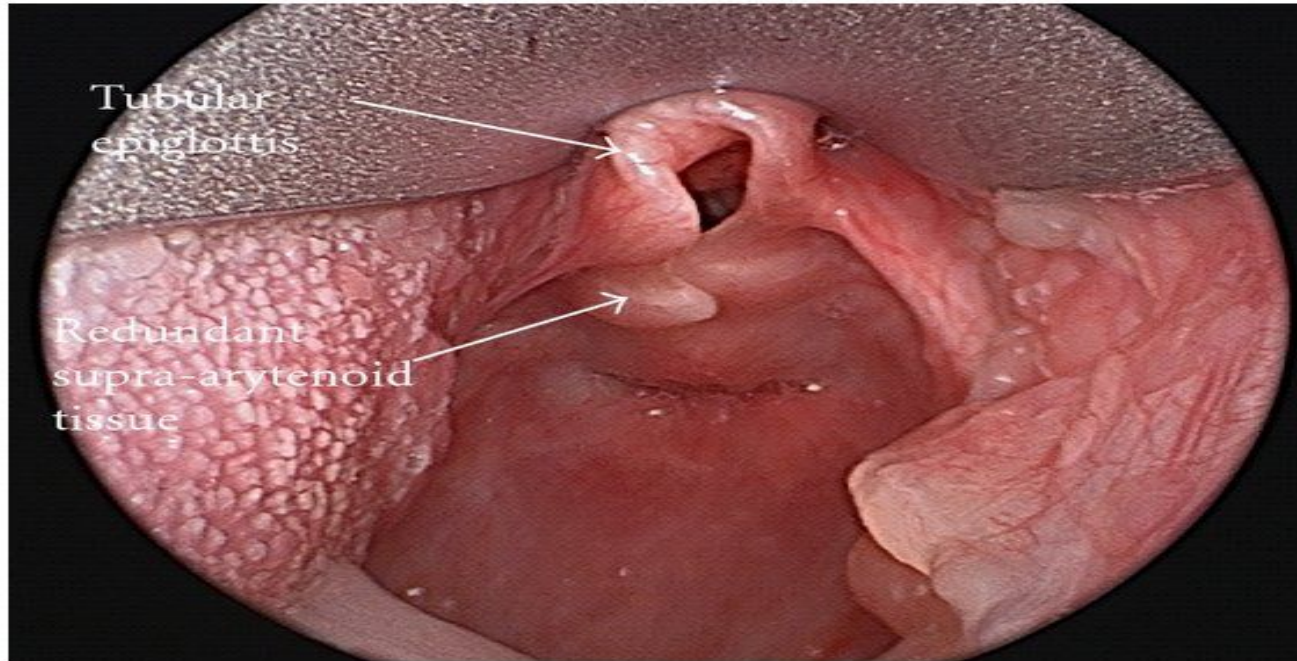
U/L SLN (ADDUCTOR) PALSY

B/L SLN(ABDUCTOR) PALSY

LARYNGOMALACIA

M/C CONGENITAL ANOMALY OF LARYNX

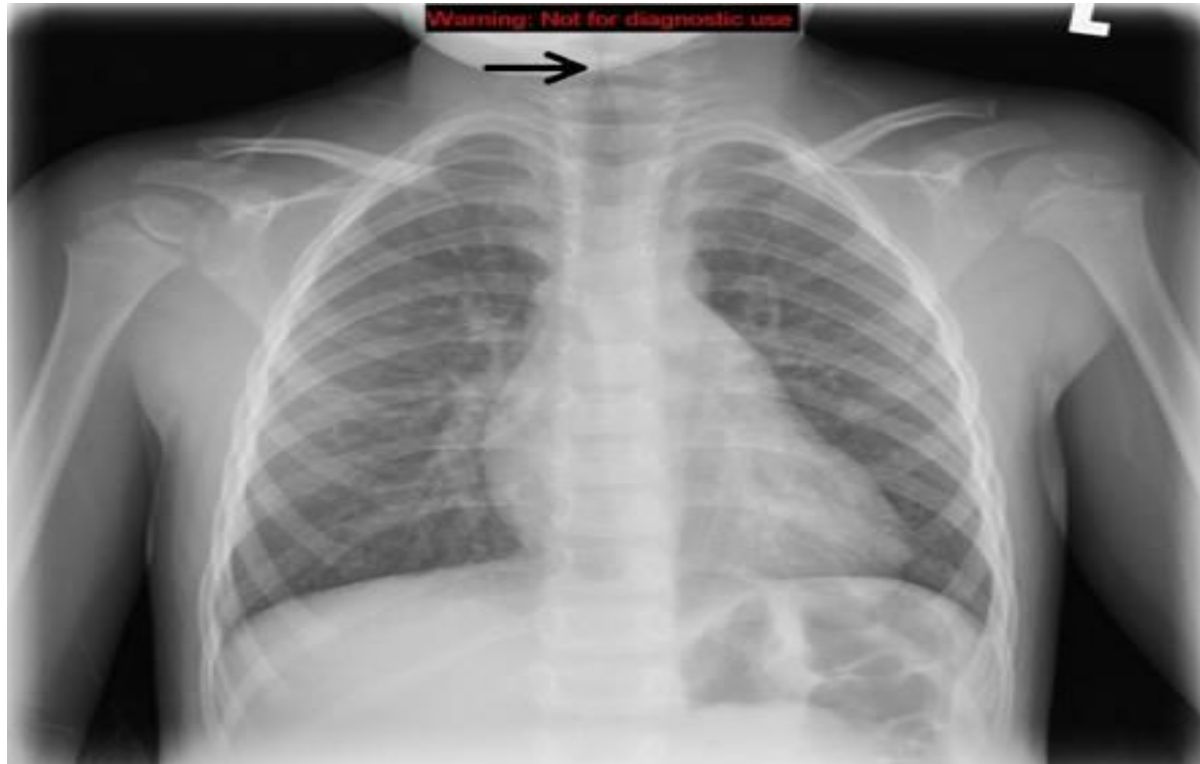
STRIDOR DECREASES ON PRONE POSITION



EPIGLOTTITIS



CROUP- STEEPLE SIGN



SOME POINTS TO DISCUSS AT LAST

LARYNGEAL CARCINOMA

LARYNGOCELE

LARYNGEAL PAPILLOMA

COTTON MEYER OF SUBGLOTTIC STENOSIS

VOICE REHAB AFTER LARYNGECTOMY

TRACHEOSTOMY